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An interpretable machine learning framework for predicting next day firm level market stress in the Dhaka Stock Exchange

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Abstract

Frequent market instability and the lack of rigorously validated forecasting frameworks pose significant challenges for predicting market stress in the Dhaka Stock Exchange (DSE). This study proposes a leakage-safe machine learning framework for next-day market stress prediction using historical trading data from 2008 to 2022. Technical indicators representing trend, momentum, volatility, and volume are extracted, while market conditions are classified into three categories: *Normal*, *High-Volatility*, and *Crash*. Seven models, namely Logistic Regression, Random Forest, XGBoost, Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Hidden Markov Model (HMM), and GJR-GARCH, are evaluated using expanding-window walk-forward cross-validation with embargo periods. Random Forest achieves the best crash prediction performance with a crash PR-AUC of 0.1697 ± 0.0134, followed by XGBoost and LSTM. At the same time, the Friedman test confirms statistically significant differences among the models ($\chi^2 = 19.11$, $p = 0.004$). Permutation Importance and SHAP identify Bollinger Bandwidth and rolling volatility as the most influential predictors. The proposed framework demonstrates that leakage-safe and interpretable machine learning can effectively support market stress prediction and risk-aware decision-making in the DSE.

Keywords Market stress prediction, Dhaka Stock Exchange, Machine learning, Financial time series analysis, Feature importance, Statistical testing

1 Introduction

Financial markets in emerging economies are inherently susceptible to episodes of heightened volatility, systemic stress, and abrupt price dislocations that can inflict severe economic harm on investors, institutions, and regulators alike [1, 2]. The DSE, as the principal securities marketplace of Bangladesh, offers a particularly compelling case for examining the dynamics of market stress and crash risk, given the series of dramatic upheavals it has experienced over recent decades. Between 2008 and 2022, the DSE was shaped by a confluence of economic, political, and global shocks that repeatedly tested

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the market's resilience and participants' confidence. The Global financial crisis of 2007–2008 curtailed international capital flows into emerging markets and suppressed investor sentiment across the region [3]. This was followed by a rapid speculative surge during 2009–2010 that inflated asset prices beyond fundamental values [4], culminating in the catastrophic Bangladesh share market crash of December 2010 to February 2011, during which extreme volatility forced regulators to temporarily suspend trading and triggered widespread investor protests [4]. The prolonged aftermath of that crash, spanning 2011 to 2013, saw a sustained erosion of market confidence and a marked contraction in trading activity [5].

Subsequent years brought further instability. Political unrest and election-related tension during 2013–2014, banking sector liquidity tightening in 2015–2016, and the Bangladesh Bank cyber robbery of 2016 cumulatively undermined systemic trust in the financial infrastructure [6]. A prolonged bear market in 2019 and political uncertainty surrounding the 2018 national elections continued to suppress market performance [7]. Most consequentially, the COVID-19 pandemic in 2020 precipitated a global economic disruption of historic proportions, resulting in an unprecedented 66-day shutdown of the DSE from March 26 to May 30, 2020, fundamentally altering the risk–return dynamics of the market [8]. A temporary liquidity-driven rally in 2021 was followed by renewed corrections, and by 2022, global inflationary pressures, the energy crisis triggered by the Russian invasion of Ukraine, and a domestic dollar shortage further amplified investor uncertainty and market instability [9].

The foregoing chronicle of crises underscores a fundamental limitation of conventional stock market research, which has historically concentrated on predicting future price levels or directional movements [10]. While such studies offer valuable insights into return dynamics, they do not directly address the problem that most concerns market participants during periods of heightened uncertainty: the identification of when the market is likely to enter a state of stress, extreme volatility, or crash. Regulators charged with maintaining orderly market conditions, investors seeking to protect their portfolios, and policymakers designing circuit-breaker mechanisms all require a predictive tool that classifies the qualitative state of the market rather than merely forecasting its quantitative trajectory [1, 2, 11].

The academic literature on stock market analysis has evolved considerably in recent years, with machine learning (ML) and deep learning (DL) approaches increasingly supplanting traditional econometric methods in predictive tasks [1, 10, 12–19]. Studies conducted on various global and emerging markets have demonstrated the superior performance of ensemble models, gradient-boosted trees, and long short-term memory (LSTM) networks over conventional approaches such as ARIMA and GARCH-family models [20–23]. In the context of the DSE specifically, a growing body of research has applied statistical methods, neural networks, support vector machines, and hybrid architectures to forecast stock prices and market directions [24–26]. However, these studies have predominantly framed the forecasting problem as either a regression task targeting closing price levels or a binary classification task distinguishing up from down movements. The more nuanced and practically consequential challenge of classifying market days into distinct risk states, Normal conditions, high-volatility episodes, and crash events has received comparatively little attention in the literature, particularly in the context of frontier markets such as Bangladesh [11, 21, 22, 27].

Furthermore, while ensemble learning approaches have been shown to enhance predictive accuracy by aggregating the outputs of heterogeneous base learners through stacking architectures [28–30], their application to market stress classification in emerging markets remains underexplored. The integration of explainable AI (XAI) techniques, into the prediction framework addresses this transparency requirement and constitutes a meaningful contribution to the emerging literature on responsible financial AI [31, 32]. Against this background, the main contributions of this study are clearly defined and systematically presented in the following section. These contributions summarize the key methodological, experimental, and analytical advancements made in this work. They are outlined as follows:

(a) A comprehensive review of the literature on stock market prediction, volatility modeling, and market stress classification, with particular emphasis on emerging and frontier markets, including the Dhaka Stock Exchange (DSE).
(b) The first application of a three-class market stress classification framework: *Normal*, *High-Volatility*, and *Crash*, to the Dhaka Stock Exchange, using historical market data and robust statistical labeling to capture varying levels of market stress.
(c) The development and comparative evaluation of multiple machine learning and time-series models, including Logistic Regression, Random Forest, XGBoost, LSTM, GRU, HMM, and GJR-GARCH, using a leakage-safe temporal validation framework for market stress prediction.
(d) The integration of model interpretability techniques, including Permutation Importance and SHAP analysis, to identify the most influential market indicators and improve the transparency of model predictions for practical financial decision-making.
(e) A comprehensive empirical evaluation incorporating class-imbalance handling, calibration assessment, regime-specific robustness analysis, statistical significance testing, and baseline comparisons to ensure the reliability and generalizability of the proposed framework.

2 Related work

2.1 Studies on the Dhaka Stock Exchange

The scholarly investigation of the Dhaka Stock Exchange encompasses a diverse range of methodological traditions, spanning classical econometric modeling, behavioral finance, and modern machine learning. A foundational strand of this literature examines the informational efficiency of the market. One study analyzed daily closing price data of the DSEX, DSE30, and DSES indices from 2015 to 2023 using both parametric and non-parametric tests, finding evidence of autocorrelation, non-stationary returns, and predictable price patterns that indicate systematic deviations from weak-form market efficiency [33]. The study attributes such inefficiencies to structural characteristics of the market, including low liquidity, limited regulatory oversight, and behavioral biases prevalent among emerging market participants, while acknowledging that behavioral finance perspectives were not fully incorporated and should be addressed in future work.

The impact of the COVID-19 pandemic on the DSE has attracted considerable scholarly attention, reflecting the singular significance of the market's 66-day shutdown in 2020. Multiple studies examined how this disruption altered market behavior and risk–return dynamics by comparing pre-lockdown and post-reopening return patterns

across individual stocks and sectoral aggregates [34, 35]. These investigations collectively highlight the structural fragility of a frontier market when confronted with a global health emergency, identifying significant increases in volatility and distributional shifts in returns that persisted well beyond the initial shock period. A complementary study employing the GJR-GARCH (1,1) model found that COVID-19 significantly elevated market volatility, while simultaneously documenting a positive effect on stock returns attributable to factors such as lower interest rates, improved remittances, and export recovery[36]. These apparently paradoxical findings underscore the non-linear and asymmetric nature of market responses to systemic crises.

Investor behavior during crisis periods has also been examined through the lens of herding. A study applying Cross-Sectional Standard Deviation (CSSD) and Cross-Sectional Absolute Deviation (CSAD) models to DSE 30 index data from 2013 to 2023, supplemented by DSE 20 data from 2007 to 2012, documented significant herding behavior across multiple market phases, including the pre-crash, crash, pre-COVID, COVID, and post-COVID periods [37]. The findings reveal that herding is more pronounced in small and medium-capitalization stocks, during high-volume trading sessions, and in both bullish and bearish market environments, suggesting that investor rationality assumptions embedded in standard pricing models are systematically violated in the DSE context.

Volatility dynamics at the firm level have been explored using GARCH models applied to daily closing prices of selected DSE-listed companies over the period January 2018 to June 2023 [38]. The results reveal non-stationary return distributions with asymmetric skewness and pronounced kurtosis, rapid price fluctuations, particularly in banking and consumer goods sectors, and a declining volatility trend in the energy sector over recent years. The study recommends the use of longer historical series to improve the robustness of volatility forecasting, a limitation that motivates the multi-year empirical window adopted in the present study. The performance of closed-end mutual funds listed on the DSE between 2015 and 2023 has also been assessed using panel data methods incorporating ROI, standard deviation, beta, and Sharpe ratio, revealing heterogeneous fund behavior under pandemic stress and recovery conditions [39].

The macroeconomic determinants of stock returns on the DSE have been studied using ARDL and VAR frameworks applied to data spanning July 2010 to December 2021, with findings indicating that stock returns are negatively affected by inflation and broad money supply while being positively influenced by foreign currency reserves [40]. Notably, Granger Causality tests yielded no evidence of long-term causal effects from these macroeconomic variables, a result consistent with findings from both developed and developing market studies. This absence of robust macroeconomic predictors further motivates the feature engineering approach adopted in the present study, which derives predictive variables directly from market microstructure rather than from external economic indicators.

2.2 Machine learning approaches for stock market prediction on the DSE

The application of machine learning techniques to stock price forecasting on the DSE has grown substantially in recent years. An early contribution applied Support Vector Machines (SVM) to predict daily closing prices of five DSE-listed companies using data from January 2017 to August 2019, demonstrating that SVM outperforms both ARIMA

and exponential smoothing methods in predictive accuracy for non-stationary financial time series [24]. Similarly, a study employing hybrid rough set and neural network models to predict optimal buy and sell decisions for DSE shares found that the combined model achieves superior prediction precision relative to either constituent approach alone, recommending the exploration of advanced reduction techniques and genetic algorithms as avenues for further improvement [41]. These early hybrid studies establish the general principle, reinforced throughout the literature, that model combination consistently outperforms single-model approaches in financial forecasting contexts [1, 14–16, 23].

Subsequent research applied Artificial Neural Networks (ANN) and the Adaptive Neuro-Fuzzy Inference System (ANFIS) to predict DSE closing prices, with ANFIS demonstrating superior performance over ANN, particularly in capturing the non-linear mapping between historical price inputs and future price outputs [42]. A related study evaluating four training algorithms for a multilayer feedforward backpropagation ANN, Levenberg-Marquardt, Bayesian Regularization, Scaled Conjugate Gradient, and Quasi-Newton, across fifteen DSE companies over six years identified the Levenberg-Marquardt algorithm as achieving the highest predictive accuracy at 95.64% with the lowest mean prediction error [25]. These results highlight the sensitivity of neural network performance to the choice of optimization procedure, an insight that motivates the systematic hyperparameter evaluation conducted in the present study.

The deployment of Long Short-Term Memory (LSTM) networks for DSE price forecasting has also been demonstrated, with studies reporting improved prediction performance over benchmark methods when the LSTM model is iteratively refined on historical data and evaluated using Root Mean Square Error [43]. A comparative study benchmarking ARIMA, SVM, and LSTM models on DSE and Chittagong Stock Exchange (CSE) data before and after the COVID-19 pandemic found that LSTM consistently achieves the highest prediction accuracy across both exchanges and both periods, reinforcing the suitability of recurrent architectures for financial time series with temporal dependencies [20].

More recent contributions have adopted broader feature engineering and ensemble strategies. A comprehensive study developed tree-based classifiers, Random Forest, XGBoost, and LightGBM, trained on 25 years of DSE historical data from 2000 to 2025, incorporating lagged prices, returns, volatilities, momentum factors, and calendar features, achieving cross-validated accuracies above 85% and deploying the LightGBM model in a production-ready web application for real-time trend prediction [21]. Feature importance analysis in that study identified rolling volatilities and momentum indicators as the most predictive variables, consistent with the feature set constructed in the present study [2, 13, 18]. A separate investigation utilizing big data analytics and machine learning integrated technical indicators with sentiment analysis from news articles and social media to improve the real-time accuracy of DSE market predictions, emphasizing the growing importance of unstructured data in financial forecasting [44].

The use of machine learning models with trading indicators for predicting DSE price directions has further been explored by combining historical price and volume data with technical signals, demonstrating promising performance in monitoring market trends [45]. A study examining machine learning techniques across supervised learning, deep learning, and hybrid architectures, including LSTM, SVM, and ANN combined with

MACD and RSI indicators, concluded that hybrid models provide the highest precision and that incorporating sentiment analysis enhances adaptability to market dynamics [12]. These findings collectively establish that ensemble and hybrid architectures represent the state of the art for DSE prediction tasks, motivating the meta-learning framework proposed in the present study.

2.3 Forecasting and trend analysis in global markets

Beyond the DSE-specific literature, a substantial body of research has demonstrated the advantages of ensemble learning approaches for stock market classification in global and regional markets. A stacked ensemble model combining XGBoost, LightGBM, and SVR on the NIFTY-50 dataset achieved an R^2 of 0.9997, confirming the robustness of stacking architectures over individual models while noting interpretability as an unresolved challenge [28]. An ensemble framework combining Logistic Regression, Random Forest, Decision Tree, SVM, KNN, AdaBoost, and Lasso-LSTM achieved 95.8% accuracy and an AUC of 0.97 on S&P 500 data, confirming that ensemble methods consistently outperform single classifiers while identifying generalization to emerging markets as a critical gap [29]. A related study demonstrated that Random Forest achieved the highest accuracy at 91.27% among nine models, emphasizing that financial simulation incorporating risk and maximum drawdown measures is necessary for comprehensive evaluation [30].

The specific problem of predicting large stock price crashes has been addressed in a study forecasting maximum drawdown events across 2812 Korean stocks from 2015 to 2024, where CatBoost achieved the best performance with an AUC of 0.841 and PR-AUC of 0.320 [22]. SHAP analysis identified implied volatility and wavelet features as the most predictive variables, confirming the general superiority of tree-based models over deep learning on tabular financial data, an insight that directly informs the base learner selection in the present study.

Machine learning applications in Indian and Malaysian markets further support the value of combining technical indicators with supervised learning models. LSTM achieved the highest accuracy (85.3%) and Sharpe ratio (1.45) when integrated with SMA, EMA, Stochastic RSI, and price-volume indicators [46], while KNN outperformed logistic regression and LDA augmented with SMA and EMA on Indian equity data [47]. Logistic Regression with PPO, MACD, RSI, Bollinger Bands, ROC, and DX signals achieved 96.467% directional accuracy, demonstrating that simple classifiers can be highly competitive with rich feature engineering [48]. In the Vietnamese market, LSTM combined with SMA, MACD, and RSI achieved 93% accuracy, though forecasting challenges persisted during strong price fluctuations [49]. Cross-market deep learning incorporating stock, bond, and foreign exchange data significantly improved Chinese stock forecasts through a GRU-Arbitrage system [50], reinforcing the broader principle that multi-feature dynamics enrich the predictive information set [17, 19].

The role of explainable AI in financial market prediction has received growing attention as practitioners and regulators demand greater transparency from complex machine learning models. A study re-evaluating 166 asset pricing characteristics using XAI confirmed that momentum and trading-based features hold the highest predictive power, with long-short portfolio analysis validating their economic significance [31]. Sentiment analysis through the ESAN model achieved 89% accuracy in investor sentiment

Table 1 Summary of methodology, dataset, features, and key findings across the reviewed studies on stock market prediction and volatility modeling

References	Dataset	Features	Methodology	Findings
Shikder et al. [38]	Daily closing prices of Arab Bangladesh Bank Limited, Bata Shoe Company, and Jamuna Oil Company Limited listed on the Dhaka Stock Exchange, covering the period from January 2018 to June 2023	Return skewness and volatility clustering patterns derived from daily closing prices	GARCH-based statistical and mathematical modeling of firm-level return volatility	The study finds that the daily return series of the selected firms are non-stationary and exhibit both positive and negative skewness, with Arab Bangladesh Bank Limited and Bata Shoe Company showing markedly higher and more erratic volatility than Jamuna Oil Company Limited, whose volatility instead follows a gradually declining trend over the study period
Golder et al. [36]	Dhaka Stock Exchange market indices spanning February 2014 to September 2021, incorporating the COVID-19 pandemic period	Pandemic-related indicators, investor fear sentiment, prevailing interest rates, and remittance inflows	Asymmetric volatility modeling using the GJR-GARCH(1,1) specification	The pandemic is found to have a statistically significant positive association with market volatility as well as with returns, an outcome attributed to investor fear, favorable valuations, lower interest rates, and improved remittance inflows, while all examined indices display mean-reverting behavior
Islam et al. [24]	Daily closing prices of five Dhaka Stock Exchange-listed companies from January 2017 to August 2019	Non-stationary daily closing price series used to train and evaluate predictive models	Support Vector Machine regression benchmarked against ARIMA and single and double exponential smoothing	The Support Vector Machine model is reported to produce predictions closely matching actual prices and to yield the lowest forecasting error among the compared methods, indicating strong predictive capability for this market
Ansary [45]	Historical price and volume series used for training several predictive models	Technical trading indicators derived from historical price and volume behavior	Comparative evaluation of multiple machine learning models for stock price direction prediction	The study reports encouraging results across the applied models while emphasizing that inherent market complexity, information asymmetry, and behavioral factors continue to limit the achievable accuracy of direction prediction
Sinha et al. [20]	Closing price data from both the Dhaka Stock Exchange and the Chittagong Stock Exchange, compared across pre-pandemic and post-pandemic periods	Closing price series segmented into pre-COVID and post-COVID subsets for comparative analysis	Comparative modeling using ARIMA, Support Vector Machine, and Long Short-Term Memory networks	The Long Short-Term Memory model is found to provide better predictive accuracy than both ARIMA and Support Vector Machine for each exchange and across both pandemic sub-periods

Table 1 (continued)

References	Dataset	Features	Methodology	Findings
Hasan et al. [21]	Twenty-five years of Dhaka Stock Exchange historical data spanning January 2000 to February 2025	Lagged prices and returns, rolling means and volatilities across multiple horizons, momentum indicators, exponential moving averages, volume lags, and calendar-based features	Binary trend classification using Random Forest, XGBoost, and LightGBM under a strict time series cross-validation scheme, combined through a soft voting ensemble	Gradient boosting methods, particularly LightGBM, are found to offer the strongest balance of accuracy, computational efficiency, and stability, with rolling volatility and momentum features identified as the most influential predictors, and the best performing model deployed as a real-time web application
Banik et al. [41]	Historical Dhaka Stock Exchange price data used to identify optimal buy and sell timing	Price-based features supporting buy and sell decision optimization	Comparative evaluation of a Rough Set model, a Neural Network model, and a hybrid combination of the two	The hybrid Rough Set and Neural Network model is shown to outperform either standalone approach in identifying optimal trading decisions, suggesting that combining feature reduction with neural modeling improves investment timing guidance
Islam et al. [12]	Dhaka Stock Exchange time series data synthesized across multiple prior studies	Moving Average Convergence Divergence and Relative Strength Index technical indicators, alongside sentiment and big data inputs discussed in the reviewed literature	Comparative review of Long Short Term Memory networks, Support Vector Machines, ensemble models, and hybrid Artificial Neural Network approaches incorporating technical indicators	The review concludes that machine learning techniques generally outperform traditional fundamental and technical analysis for short-term forecasting, with hybrid models combining neural networks and technical indicators offering the greatest precision, although limited data quality and insufficient macroeconomic integration remain persistent challenges
Khan et al. [25]	Opening, high, and low prices along with shares traded for fifteen Dhaka Stock Exchange-listed companies over six years	Opening, high, and low price series together with trading volume for each company	Artificial Neural Network with multilayer feedforward backpropagation, trained using Levenberg Marquardt, Bayesian Regularization, Scaled Conjugate Gradient, and Quasi Newton algorithms	The Levenberg-Marquardt network is found to substantially outperform the other training algorithms in prediction accuracy and error reduction, establishing it as the most suitable approach among those tested
Kamwele Mutinda et al. [13]	Six major global indices, namely the FTSE 100, Shanghai Stock Exchange Composite, S&P 500, DAX, NYSE Composite, and Hang Seng Index, were evaluated under a hold-out split	Intrinsic mode functions obtained through CEEMDAN and further refined through VMD, grouped by frequency band using sample entropy and K-means clustering	A hierarchical CEEMDAN SE VMD BiGRU Ridge hybrid framework combining bidirectional gated recurrent units for high and medium frequency components with ridge regression for low frequency and residual components	The proposed framework is shown to outperform thirteen benchmark models across all six indices, with superiority confirmed through Modified Diebold Mariano and Model Confidence Set tests, and robustness demonstrated across multiple business cycles and contrasting market regimes

Table 1 (continued)

References	Dataset	Features	Methodology	Findings
Kamwele Mutinda et al. [14]	Six major global equity indices, namely the FTSE 100, Shanghai Stock Exchange, S&P 500, DAX, NYSE, and Hang Seng Index	Rolling window technical indicators filtered through a novel Stability Enhanced Dynamic Thresholding feature selection procedure	Eight classifiers, including Logistic Regression, Random Forest, Support Vector Machine, XGBoost, Artificial Neural Network, LSTM, GRU, and CNN, were evaluated with the proposed feature selection method against Plain, LASSO, Boruta, and RFECV baselines	The proposed feature selection approach is found to consistently outperform all baseline methods across indices and classifiers, with SHAP analysis revealing that momentum and oscillator indicators dominate feature importance, and trading simulations confirming improved returns and reduced risk relative to baseline strategies
Kamwele Mutinda et al. [15]	Airtel stock price data evaluated under multiple train-test partitioning schemes	Intrinsic mode functions obtained through variational mode decomposition, separated into high-frequency and low-frequency components	A hybrid VMD GARCH LSTM framework in which GARCH captures volatility in high-frequency components before LSTM modeling, while low-frequency components are modeled directly by LSTM	The hybrid model is reported to consistently outperform fourteen benchmark architectures across all data partitions, with statistical superiority confirmed through Diebold Mariano testing, underscoring its stability and reliability for price forecasting
Kamwele Mutinda et al. [23]	The DAX market index evaluated across multiple training and testing split ratios	Intrinsic mode functions and residuals obtained through CEEMDAN decomposition, classified into low, medium, and high frequency components	A hybrid CEEMDAN LSTM BPNN model applying LSTM to medium and high frequency components and a backpropagation neural network to low frequency components	The hybrid model is shown to significantly outperform ARIMA, RNN, and several other benchmark and CEEMDAN-based architectures, with its superiority confirmed through Diebold Mariano and Model Confidence Set testing across multiple evaluation metrics
Kamwele Mutinda et al. [16]	Airtel stock price data obtained from Yahoo Finance, spanning June 2019 to May 2024	Statistical properties extracted through GARCH modeling combined with historical price data	Hybrid GARCH LSTM, GARCH GRU, and GARCH Transformer models benchmarked against their standalone counterparts	The GARCH LSTM hybrid model is found to significantly outperform standalone LSTM, GRU, and Transformer models, demonstrating the value of integrating volatility modeling with deep learning architectures for African market price prediction
Kamwele Mutinda et al. [1]	Realized volatility series for four Chinese stocks, namely 600519.SS, 601398.SS, 601857.SS, and 601988.SS	Intrinsic mode functions obtained through variational mode decomposition optimized by signal-to-noise ratio, classified into high and low frequency segments using sample entropy and t-tests	A hybrid VMD GARCH DNN ensemble in which GARCH models high frequency components before feeding them into deep neural network architectures such as LSTM, GRU, BiGRU, and BiLSTM, while low frequency components are modeled directly by these networks	The hybrid ensemble is shown to outperform standalone deep neural network models across all evaluation metrics for both ten-day and twenty-two-day forecasting horizons, indicating robust performance for realized volatility prediction over multiple investment horizons

Table 1 (continued)

References	Dataset	Features	Methodology	Findings
Khalil [26]	A publicly available insurance claims dataset used to construct fraud detection models	Discretized and resampled features generated through combinations of feature discretization, feature selection, and data resampling strategies	An integrated pipeline combining feature discretization, recursive feature elimination, oversampling, and Random Forest classification, compared across multiple scenario combinations	The combined discretization, feature elimination, and oversampling scenario using Random Forest achieves the strongest fraud detection performance, though the absence of proprietary economic and demographic variables limits comprehensive cost-benefit and fairness analysis
Gunduz [17]	Hourly data for eight banking stocks listed on Borsa Istanbul	Own stock technical features, features reduced through Variational Autoencoders, and features incorporating other correlated banking stocks	Linear models, LSTM with attention mechanisms, and LightGBM ensemble models trained on four different feature configurations, with dimensionality further reduced through Recursive Feature Elimination	The LSTM with attention model trained on own stock features achieves the strongest overall performance, while the Variational Autoencoder reduced feature set achieves comparable results using substantially fewer input features, demonstrating the value of dimensionality reduction for hourly direction prediction
Hu et al. [11]	Stock return series labeled into up, sideways, and down trend categories	Twenty-four technical indicators used as explanatory variables within a group penalized modeling framework	Group LASSO, SCAD, and MCP penalized trinomial logit dynamic models estimated through a group coordinate descent algorithm and evaluated against deep learning and machine learning baselines	The proposed group penalized models are found to outperform their non-grouped penalized counterparts as well as several deep learning and machine learning benchmarks in three-class trend prediction, while also requiring substantially less computation time than the non-grouped penalized alternatives
Shahzad et al. [27]	Firm-level data spanning multiple life cycle stages within emerging market economies	Asset liquidity constraints, idiosyncratic risk, and cash flow risk indicators associated with each life cycle stage	Empirical regression-based analysis examining how life cycle stage moderates the relationship between liquidity, risk, and innovation investment	Firms are found to behave differently across life cycle stages, investing more heavily in input innovation during introduction, growth, and decline stages under greater asset liquidity and volatility, while such investment diminishes considerably during the mature stage
Miao et al. [18]	Nine stock datasets drawn from the liquor and spirits sector	Technical indicators organized into a nested feature subset structure combining backward elimination with weighted relevance scoring	A sparse Support Vector Machine framework integrating Recursive Feature Elimination with a weighted ReliefF filtering algorithm	The proposed sparse Support Vector Machine approach is shown to outperform baseline sparse and standard Support Vector Machine models in both accuracy and F-test performance, while also producing more compact and less redundant feature subsets

Table 1 (continued)

References	Dataset	Features	Methodology	Findings
Omole et al. [19]	Bitcoin on-chain and price data used for direction prediction and trading strategy evaluation	Features selected through Boruta, genetic algorithm, and light gradient boosting machine-based selection methods to address high dimensionality	Comparative evaluation of a CNN LSTM hybrid model, a long- and short-term time series network, a temporal convolutional network, and an ARIMA benchmark, combined with backtesting of multiple trading strategies	The combination of Boruta feature selection with the CNN LSTM model is found to achieve the strongest predictive performance among the compared architectures, and when applied to a long and short trading strategy, the resulting predictions are shown to generate substantial annual returns
Yan et al. [2]	Twelve selected American financial sector stocks	Historical volatility data used to construct and evaluate quantitative investment strategies	Three time series models combined with six machine learning models, namely K nearest neighbors, AdaBoost, CatBoost, LightGBM, XGBoost, and Random Forest, forming eighteen fusion models	The fusion models are found to enhance volatility forecasting precision, and when integrated with quantitative investment principles, the resulting strategy is shown to generate favorable average annualized returns across the examined financial stocks

identification and 79% in emotion recognition, establishing that interpretability and sentiment integration are substantive enhancements to financial prediction models [32].

The broader literature on stock market volatility and uncertainty prediction offers additional methodological context. A MISO framework tuned by Particle Swarm Optimization demonstrated that nonlinear relationships between uncertainty indicators and market volatility can be effectively captured through system identification methods, outperforming individual models and traditional econometric approaches [51]. A sigmoid-based deep learning stability model further demonstrated the ability to adapt to abrupt market fluctuations by iteratively adjusting based on prediction errors [52]. A comprehensive review identified persistent limitations, including single-algorithm reliance, neglect of textual data, and poor daily forecasting accuracy, gaps that directly motivate the multi-model stacking architecture, comprehensive feature engineering, and Lime-based interpretability incorporated in the present study [10]. A meta-regression of many papers further highlighted the growing complexity of the risk landscape as ecological uncertainty increasingly influences corporate performance and stock market behavior [53].

2.4 Gap analysis

Despite the growing body of literature on stock market prediction, several critical gaps remain unaddressed, particularly in the context of frontier markets such as the Dhaka Stock Exchange. presents a structured summary of the referenced studies, highlighting the key features, techniques employed, and findings that collectively motivate the proposed framework. Existing DSE-focused studies have predominantly framed the forecasting problem as either a regression task targeting closing price levels or a binary classification distinguishing upward from downward movements [20, 24, 41], while largely relying on single-model architectures such as SVM, ANN, LSTM, or GARCH

variants [12, 25, 36, 38]. Feature sets in prior work have been narrow, often omitting intraday volatility measures, high–low spreads, volume dynamics, and rolling momentum indicators that are particularly informative during stress episodes [20, 21, 45]. Moreover, none of the existing DSE studies have attempted to classify market conditions into distinct risk states: Normal, High-Volatility, and crash, a formulation that is arguably more actionable for investors and regulators than point-in-time price forecasts.

Beyond the DSE-specific literature, broader studies on ensemble learning approaches have demonstrated superior predictive performance over single-model methods [28, 29], yet their application to frontier market stress classification remains unexplored. A persistent and widely acknowledged limitation across the global literature is the absence of model interpretability, with deep learning and ensemble models frequently criticized for operating as black-box systems that offer little transparency to end-users [32, 50, 51]. To the best of our knowledge, no previous study has jointly explored multi-class market stress classification, machine learning techniques, and feature importance-based interpretability within a unified framework for the Dhaka Stock Exchange, leaving a significant research gap that this study aims to address (Table 1).

To address this research gap, the present study proposes a leakage-safe, data-driven machine learning framework for classifying market stress and crash risk in the Dhaka Stock Exchange (DSE). Unlike prior studies that primarily focused on price prediction or binary market direction forecasting, the proposed framework adopts a three-class market stress taxonomy comprising Normal days, High-Volatility days, and Crash days, providing a more comprehensive characterization of market conditions. The framework integrates robust feature engineering, multiple machine learning and time-series models, rigorous temporal validation, and explainability techniques to improve predictive performance and model transparency. Furthermore, the proposed approach is evaluated across historically significant market regimes and complemented with industry-wise risk profiling, offering actionable insights for investors, portfolio managers, policymakers, and financial regulators in frontier market environments.

3 Proposed methodology

The proposed framework predicts market stress conditions in the Dhaka Stock Exchange (DSE) using a leakage-safe machine learning pipeline. Historical market data (2008–2022) undergoes data validation, preprocessing, and feature engineering to extract informative trend, momentum, volatility, and volume indicators, while market stress labels are generated using robust statistical thresholds with next-day (t+1) prediction targets. The processed data is chronologically divided into training and testing sets, and multiple machine learning and time-series models, including Logistic Regression, Random Forest, XGBoost, LSTM, GRU, HMM, and GJR-GARCH, are trained using expanding-window cross-validation and optimized through systematic hyperparameter tuning. Model performance is assessed using classification, calibration, robustness, and statistical significance analyses, while permutation importance and SHAP are employed to improve model interpretability. The overall research workflow is illustrated in Fig. 1.

3.1 Data collection and validation

The dataset was initially collected from Kaggle and subsequently underwent a validation process to ensure its reliability and consistency prior to analysis [54]. The original data,

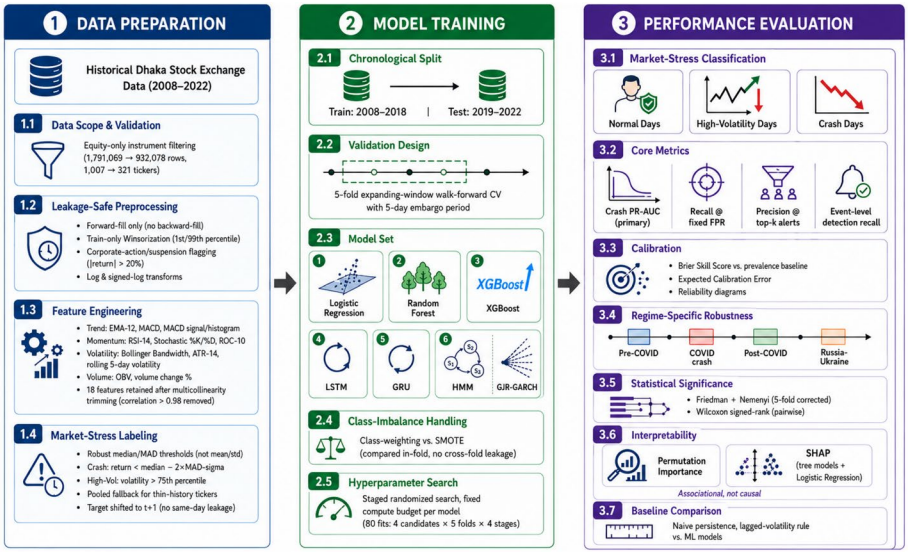


Fig. 1 Proposed leakage-safe machine learning framework for market stress prediction in the Dhaka Stock Exchange, illustrating the complete workflow from data preparation and feature engineering to model training, validation, performance evaluation, and interpretability analysis

provided in JSON format, were converted into CSV format to facilitate preprocessing, visualization, and downstream analytical tasks.

To verify the authenticity of the collected data, a cross-validation procedure was conducted using the official LankaBangla Financial Portal [55], which provides both real-time and historical trading information for the Dhaka Stock Exchange (DSE). The historical price chart corresponding to the target company was accessed through the portal, and the associated network requests were examined using the browser's developer tools to identify the underlying API endpoint responsible for retrieving historical market data. The API returned raw JSON-formatted records containing timestamped Open, High, Low, Close, and Volume (OHLCV) information, comprising 1199 total records spanning the full available history of the stock. These records were extracted and processed using a custom Python script, which converted the millisecond-resolution Unix timestamps into calendar dates and filtered the observations to match the study period (12 July 2021–29 December 2022).

After filtering, 354 trading-day records fell within the study period from both the Kaggle dataset and the officially sourced DSE data, yielding a matched sample of 354 observations (100% of the study-period records) for direct comparison. The two sources were cross-checked on a record-by-record basis with respect to trading dates, record counts, and closing prices. This comparison showed complete agreement in record count and date coverage between the two sources, with the first and last matched trading dates (18 July 2021 and 29 December 2022, respectively) identical across both datasets. Only a single discrepancy was identified, corresponding to one non-trading (halted) session on 5 May 2022, which was consistently flagged as a zero-volume/non-trading observation in both sources rather than representing a genuine data inconsistency. No missing trading dates, duplicate records, or conflicting closing-price values were found. Given this 100% agreement rate across the 354 compared records, the Kaggle dataset was considered validated, complete, and reliable, and was therefore deemed suitable for subsequent preprocessing, feature engineering, and predictive modeling.

Table 2 Definitions, constituent variables, and inclusion rationale for the 18 technical indicators retained in the final feature set after correlation-based redundancy removal

Feature name	Definition	Variables used	Rationale for inclusion
Daily return	Percentage change in closing price relative to the previous session (Eq. 7)	Close_t, Close_{t-1}	Captures short-term directional price movement, the basic unit of return-based risk
High-low spread	Absolute difference between the session's high and low prices (Eq. 8)	High, Low	Reflects raw intraday trading range as a simple dispersion measure
Intraday volatility	High-low spread normalized by closing price (Eq. 9)	High, Low, Close	Scale-free volatility measure comparable across price levels and used directly in the stress-labeling rule
Volume change (%)	Percentage change in traded volume relative to the previous session (Eq. 10)	Volume_t, Volume_{t-1}	Captures shifts in market participation and liquidity that often precede price moves
Price gap	Difference between the current session's opening price and the prior session's closing price (Eq. 11)	Open_t, Close_{t-1}	Detects overnight information shocks not reflected in the previous close
Rolling 5-day volatility	Standard deviation of daily returns over the trailing 5-day window (Eq. 12)	$r_{t-4}, \ldots, r_t$	Provides a short-horizon, smoothed volatility signal less sensitive to single-day noise
Rolling 10-day momentum	Change in closing price over a trailing 10-day window (Eq. 13)	Close_t, Close_{t-10}	Captures medium-term trend persistence beyond single-day returns
EMA_{12}	12-period exponentially weighted moving average of closing price (Eq. 14)	Close_t, $\text{EMA}_{12,t-1}$	Retained trend-following baseline after removing redundant SMA/EMA/Bollinger-band variants
MACD	Difference between EMA_{12} and EMA_{26} (Eq. 15)	EMA_{12}, EMA_{26}	Summarizes short- vs. long-term trend divergence
MACD signal	9-period EMA of the MACD line (Eq. 15)	MACD_t	Smooths MACD to generate trend-reversal reference points
MACD histogram	Difference between MACD and its signal line (Eq. 15)	MACD_t, $\text{MACD}_{\text{signal},t}$	Highlights acceleration or deceleration in trend momentum
RSI-14	14-period relative strength index based on mean gains vs. losses (Eq. 16)	Close (14-day gains/losses)	Identifies overbought/oversold conditions linked to potential reversals
%K	Position of the current close within the 14-day high-low range (Eq. 17)	Close_t, High_{14}, Low_{14}	Stochastic measure of relative price positioning within recent range
%D	3-period SMA of %K (Eq. 17)	$\%\text{K}_t$	Smoothed stochastic signal used to reduce %K noise
ROC-10	Percentage price change over a 10-day lookback (Eq. 17)	Close_t, Close_{t-10}	Alternative momentum measure independent of the volatility-based indicators
Bollinger bandwidth	Normalized width between upper and lower Bollinger Bands relative to the 20-day SMA (Eq. 18)	SMA_{20}, σ_{20}	Retained volatility-dispersion measure after removing highly correlated raw band levels
ATR-14	14-day average true range accounting for overnight gaps (Eq. 19)	High, Low, Close_{t-1}	Captures typical daily price movement including gap risk, complementing intraday volatility
OBV	Cumulative volume adjusted by the sign of the daily price change (Eq. 20)	Close_t, Close_{t-1}, Volume_t	Tracks directional volume pressure as a leading indicator of price trend strength

3.2 Data preprocessing

The dataset consists of 1,791,069 daily trading records described by 14 attributes. Monetary values are reported in **millions of Bangladeshi Taka (BDT)**, following the standard reporting convention of the Dhaka Stock Exchange (DSE), while trading activity is represented using the actual number of shares exchanged. Exploratory analysis confirmed the absence of missing values in these variables, with traded values reaching up to 4,365.43

million BDT and trading volumes up to 94,006,000 shares, indicating that the dataset is complete and suitable for subsequent analysis.

To ensure the dataset reflected genuine equity trading behavior, records were restricted to instruments classified as *Equity*, excluding fixed-income and other non-equity instrument types; this reduced the dataset from 1,791,069 rows across 1007 trading codes to 932,078 rows across 321 equity tickers. Infinite values arising from zero-division cases in the return and volatility calculations (Eq. 1) were converted to missing values and subsequently neutralized prior to any statistical fitting.

A dedicated screening step was applied to identify single-day return observations that were implausibly extreme for genuine market movements. Raw daily returns exceeding $\pm 20\%$ in a single session were flagged as suspected trading suspensions, delistings, or unadjusted corporate actions (Eq. 2). These rows were retained in the dataset, since a suspension or delisting is itself a legitimate market-stress event worth capturing, but were excluded from the statistics used to fit the labeling thresholds described below, preventing a small number of extreme observations from distorting the thresholds applied to the remaining data. This screening identified 1568 such rows (0.168% of the equity subset) prior to feature-completeness filtering, affecting a broad cross-section of tickers rather than any single stock.

Missing values introduced by rolling-window feature computations were handled using *forward fill only*, applied independently within each trading code (Eq. 3). Backward filling was deliberately excluded from the pipeline, as propagating a future observation backward in time would introduce look-ahead information into earlier records. Rows retaining missing values after forward filling, corresponding to the initial warm-up period of each stock's trading history, before sufficient historical observations existed to compute rolling-window indicators, were dropped rather than imputed, since fabricating values for a feature's genuine absence period would itself constitute a form of leakage.

Outlier capping was performed using Winsorization at the 1st and 99th percentile bounds (Eq. 4); critically, these bounds were computed using training-period data only and then applied uniformly to both the training and test partitions, preventing information about the test distribution from influencing the capping thresholds. Logarithmic and signed logarithmic transformations (Eqs. 5, 6) were subsequently applied to features exhibiting substantial skewness, improving distributional stability for model training. As these transformations are parameter-free, they were applied uniformly across the full dataset without risk of leakage.

$$x = \begin{cases} \text{NaN}, & \text{if } x = \pm\infty \\ x, & \text{otherwise} \end{cases} \tag{1}$$

$$\text{suspected_corp_action}_t = \begin{cases} 1, & |R_t| > 20\% \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

$$x_t = \begin{cases} x_{t-1}, & \text{if } x_t \text{ is NaN and a prior valid observation exists} \\ \text{dropped}, & \text{if no prior valid observation exists (warm-up period)} \end{cases} \tag{3}$$

$$x_{\text{wins}} = \begin{cases} P_1^{\text{train}}, & x < P_1^{\text{train}} \\ x, & P_1^{\text{train}} \le x \le P_{99}^{\text{train}} \\ P_{99}^{\text{train}}, & x > P_{99}^{\text{train}} \end{cases} \tag{4}$$

$$x' = \log(1 + x) \tag{5}$$

$$x' = \text{sign}(x) \log(1 + |x|) \tag{6}$$

where R_t denotes the raw daily return prior to any transformation, P_1^{train} and P_{99}^{train} denote the 1st and 99th percentile bounds computed exclusively on the training partition, and all remaining notation follows the definitions given previously.

3.3 Feature engineering

A comprehensive set of technical indicators was constructed from the underlying price and volume series to characterize short-, medium-, and trend-level market dynamics, extending the base feature set defined: daily return (Eq. 7), high-low spread (Eq. 8), intraday volatility (Eq. 9), volume change percentage (Eq. 10), price gap (Eq. 11), rolling 5-day volatility (Eq. 12), and rolling 10-day momentum (Eq. 13) [56–58].

To capture trend structure, simple and exponential moving averages (Eq. 14) were computed at multiple horizons and used to derive the Moving Average Convergence Divergence indicator (Eq. 15), which tracks the relationship between a fast and slow exponential moving average along with its associated signal line and histogram. Momentum was further characterized using the Relative Strength Index (Eq. 16), which measures the magnitude of recent gains relative to recent losses to identify overbought or oversold conditions, alongside the Stochastic Oscillator and Rate of Change (Eq. 17), which respectively position the current closing price within its recent trading range and quantify percentage price change over a fixed lookback window.

Volatility structure was captured using Bollinger Bands (Eq. 18), which express price dispersion relative to a rolling mean, and the Average True Range (Eq. 19), which accounts for gaps between sessions when measuring typical daily price movement. Finally, On-Balance Volume (Eq. 20) was computed as a cumulative volume-flow indicator that adds or subtracts each day's volume according to the direction of the price change, reflecting the directional pressure of trading activity.

$$\text{Daily Return} = \frac{\text{Closing Price} - \text{Yesterday's Closing Price}}{\text{Yesterday's Closing Price}} \times 100 \tag{7}$$

$$\text{High-Low Spread} = \text{High} - \text{Low} \tag{8}$$

$$\text{Intraday Volatility} = \frac{\text{High} - \text{Low}}{\text{Closing Price}} \times 100 \tag{9}$$

$$\text{Volume Change (\%)} = \frac{\text{Volume}_t - \text{Volume}_{t-1}}{\text{Volume}_{t-1}} \times 100 \tag{10}$$

$$\text{Price Gap} = \text{Opening Price}_t - \text{Closing Price}_{t-1} \tag{11}$$

$$\text{Rolling 5-Day Volatility} = \sqrt{\frac{1}{5} \sum_{i=t-4}^{t} (r_i - \bar{r})^2} \tag{12}$$

$$\text{Rolling 10-Day Momentum} = \text{Closing Price}_t - \text{Closing Price}_{t-10} \tag{13}$$

$$\mathrm{SMA}_n = \frac{1}{n}\sum_{i=t-n+1}^{t} \mathrm{Close}_i, \qquad \mathrm{EMA}_n = \alpha \cdot \mathrm{Close}_t + (1-\alpha)\cdot \mathrm{EMA}_{n,t-1}, \;\; \alpha = \frac{2}{n+1} \tag{14}$$

$$\begin{aligned} &\mathrm{MACD}_t = \mathrm{EMA}_{12,t} - \mathrm{EMA}_{26,t}, \quad \mathrm{MACD}_{\mathrm{signal},t} = \mathrm{EMA}_9(\mathrm{MACD}_t), \\ &\mathrm{MACD}_{\mathrm{hist},t} = \mathrm{MACD}_t - \mathrm{MACD}_{\mathrm{signal},t} \end{aligned} \tag{15}$$

$$\mathrm{RSI}_{14} = 100 - \frac{100}{1+RS}, \qquad RS = \frac{\text{mean gain}_{14}}{\text{mean loss}_{14}} \tag{16}$$

$$\%K_t = \frac{\mathrm{Close}_t - \mathrm{Low}_{14}}{\mathrm{High}_{14} - \mathrm{Low}_{14}} \times 100, \qquad \%D_t = \mathrm{SMA}_3(\%K_t), \qquad \mathrm{ROC}_{10} = \frac{\mathrm{Close}_t - \mathrm{Close}_{t-10}}{\mathrm{Close}_{t-10}} \times 100 \tag{17}$$

$$\mathrm{BB}_{\mathrm{upper}} = \mathrm{SMA}_{20} + 2\sigma_{20}, \quad \mathrm{BB}_{\mathrm{lower}} = \mathrm{SMA}_{20} - 2\sigma_{20}, \quad \mathrm{BB}_{\mathrm{width}} = \frac{\mathrm{BB}_{\mathrm{upper}} - \mathrm{BB}_{\mathrm{lower}}}{\mathrm{SMA}_{20}} \tag{18}$$

$$\mathrm{ATR}_{14} = \frac{1}{14}\sum_{i=t-13}^{t} \max\left(\mathrm{High}_i - \mathrm{Low}_i,\ |\mathrm{High}_i - \mathrm{Close}_{i-1}|,\ |\mathrm{Low}_i - \mathrm{Close}_{i-1}|\right) \tag{19}$$

$$\mathrm{OBV}_t = \mathrm{OBV}_{t-1} + \mathrm{sign}(\mathrm{Close}_t - \mathrm{Close}_{t-1}) \cdot \mathrm{Volume}_t \tag{20}$$

Following construction of the full indicator set, a correlation analysis was conducted on the training partition to identify redundant features prior to model training. Five indicators, SMA_{10}, SMA_{20}, EMA_{26}, $\mathrm{BB}_{\mathrm{upper}}$, and $\mathrm{BB}_{\mathrm{lower}}$, exhibited near-perfect correlation ($r > 0.98$) with retained features (principally EMA_{12} and $\mathrm{BB}_{\mathrm{width}}$) and were consequently removed, yielding a final feature set of 18 indicators used for model training: daily return, high-low spread, intraday volatility, volume change percentage, price gap, rolling 5-day volatility, rolling 10-day momentum, EMA_{12}, MACD, MACD signal, MACD histogram, RSI-14, $\%K$, $\%D$, ROC-10, Bollinger bandwidth, ATR-14, and OBV. Equation 2 summarizes the definition, constituent variables, and inclusion rationale for each of the retained features.

3.3.1 Market-stress labeling

A three-class target variable was constructed to classify each trading day as Normal, High-Volatility, or Crash, using stock-specific thresholds fitted exclusively on the training partition. Rather than mean and standard-deviation statistics, robust median and median-absolute-deviation (MAD) statistics (Eq. 21) were used to fit the crash threshold (Eq. 22), since the raw return distribution, even after excluding the suspected corporate-action rows identified earlier, retains heavier tails than a normal-distribution assumption can accommodate. The High-Volatility threshold (Eq. 23) was instead derived from the 75th percentile of the training-period intraday volatility distribution, consistent with the percentile-based approach used in the base labeling scheme.

$$\sigma_{\mathrm{robust}} = 1.4826 \times \mathrm{MAD}, \qquad \mathrm{MAD} = \mathrm{median}\left(|R_i - \mathrm{median}(R)|\right) \tag{21}$$

$$\tau_{\mathrm{crash}} = \mathrm{median}(R^{\mathrm{train}}) - 2\,\sigma^{\mathrm{train}}_{\mathrm{robust}} \tag{22}$$

$$\tau_{\mathrm{vol}} = Q_{0.75}\left(V^{\mathrm{train}}\right) \tag{23}$$

where R^{train} and V^{train} denote the raw daily return and intraday volatility series restricted to the training partition and to non-suspicious observations. For stocks with

Table 3 Class-wise distribution of the next-day market-stress target across the training and test partitions

Class	Train (n)	Train (%)	Test (n)	Test (%)
Normal	465,423	73.83%	217,365	73.79%
High-Volatility	129,879	20.60%	63,344	21.50%
Crash	35,054	5.56%	13,881	4.71%
Total	**630,356**	**100%**	**294,590**	**100%**

Bold values indicate the total number of observations and the corresponding overall percentage within eachdata partition

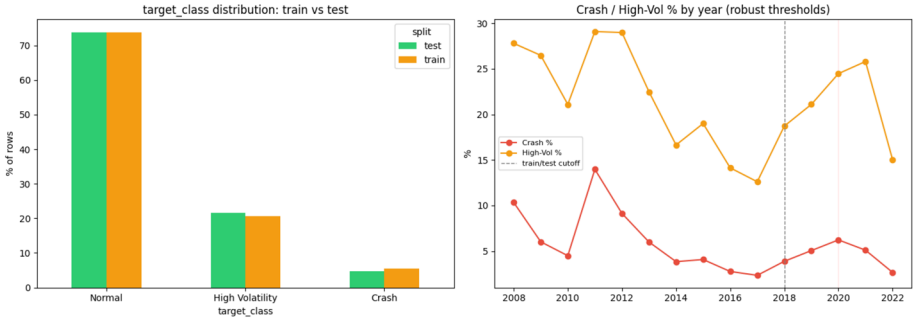


Fig. 2 Left: distribution of the three-class target (`target_class`) across the train and test splits. Right: yearly Crash % and High-Volatility % under the robust thresholding scheme, with the dashed line marking the train/test cutoff

fewer than 30 qualifying training-period observations, typically recent listings with insufficient trading history, pooled market-wide statistics, computed identically across all qualifying training-period observations, were substituted as a fallback.

Each trading day was then labeled according to a priority-based rule, in which the crash condition is evaluated first to ensure mutual exclusivity between categories:

(a) Crash Day: the raw daily return R_t falls below τ_{crash} for that stock.
(b) High-Volatility Day: the intraday volatility V_t exceeds τ_{vol}, provided the crash condition is not met.
(c) Normal Day: all remaining observations.

To ensure the labeled target reflects a genuinely predictive task rather than a same-day description, the class label was shifted forward by one trading day, such that each row's feature vector at time t is paired with the market-stress class realized at $t + 1$; the final trading day of each stock's history, for which no subsequent label exists, was dropped from the dataset.

3.4 Data analysis

Following preprocessing, feature engineering, and labeling, the final dataset comprised 924,946 observations across 321 equity tickers, partitioned into 630,356 training rows (2008–2018) and 294,590 test rows (2019–2022). The resulting three-class distribution is summarized in Table 3 and illustrated in Fig. 2, and remained closely consistent between the training and test partitions, indicating that the labeling scheme generalizes stably across the temporal split rather than being an artifact of a particular period.

A correlation analysis of the final 18-feature set on the training partition, shown in Fig. 3, confirmed that the multicollinearity trimming described above was effective: only one feature pair, the stochastic $\%K/\%D$ oscillator pair ($r = 0.930$), retained a

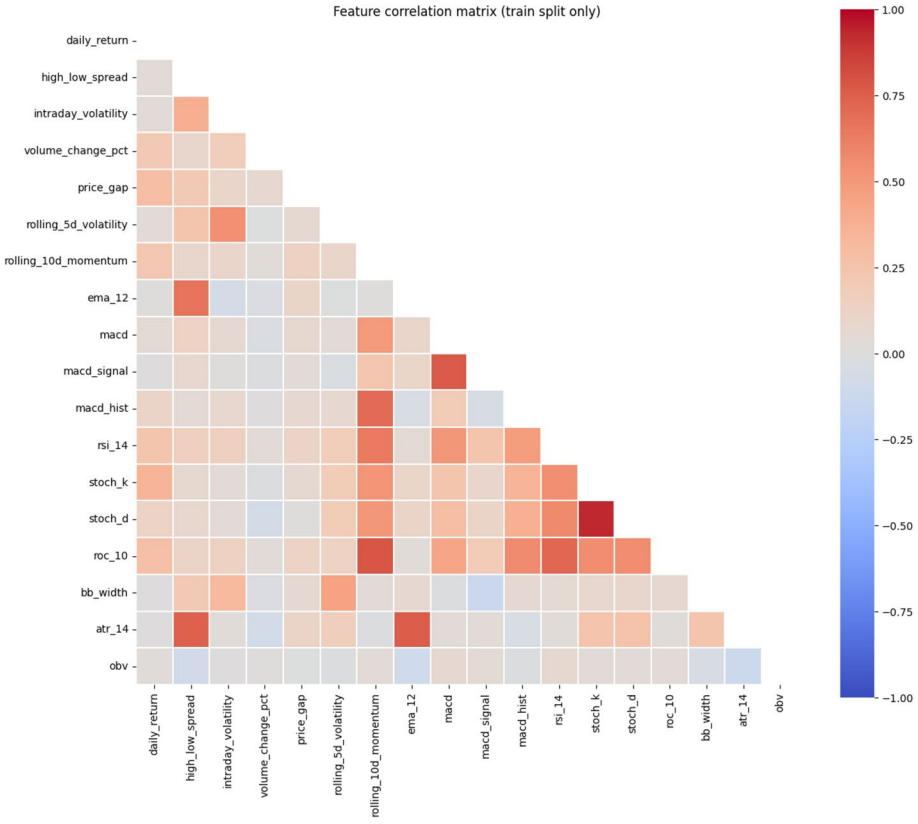


Fig. 3 Pairwise correlation matrix of the final 18-feature set, computed on the training split

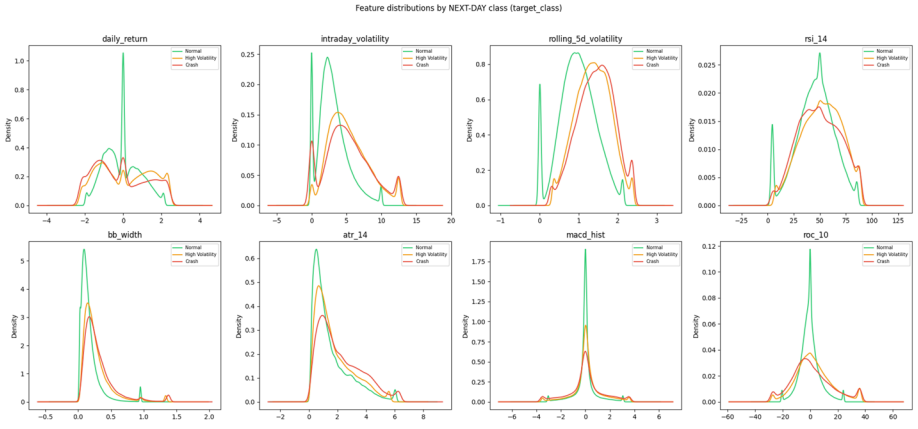


Fig. 4 Class-conditional kernel density estimates of daily return, intraday volatility, rolling 5-day volatility, RSI-14, Bollinger bandwidth, ATR-14, MACD histogram, and ROC-10, grouped by next-day market-stress class

correlation above 0.85, which reflects the intentional smoothing relationship between the two indicators rather than genuine redundancy, since $\%D$ is by construction a moving average of $\%K$.

Class-conditional kernel density plots of key indicators (daily return, intraday volatility, rolling 5-day volatility, RSI-14, Bollinger bandwidth, ATR-14, MACD histogram, and ROC-10), presented in Fig. 4, revealed visibly distinct distributions across the three market-stress classes, with Crash-day observations exhibiting the widest dispersion and heaviest left tail in daily returns, and both High-Volatility and Crash days showing

Table 4 Mean value of selected features by next-day market-stress class (training partition)

Feature	Normal	High-volatility	Crash
Daily return	− 0.074	0.115	− 0.123
Intraday volatility	3.162	5.227	4.999
Rolling 5-day vol.	1.027	1.370	1.436
RSI-14	46.780	51.698	49.444
Bollinger bandwidth	0.179	0.285	0.325
ATR-14	1.700	2.031	2.512

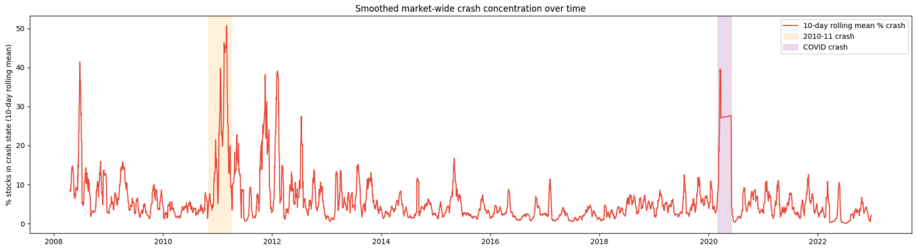


Fig. 5 Smoothed 10-day rolling mean of the cross-sectional percentage of stocks in the crash state, 2008-2022, with the 2010-2011 crash and COVID-19 crash windows highlighted

elevated intraday volatility, rolling volatility, and Bollinger bandwidth relative to Normal days. A class-conditional mean-feature summary (Table 4) further indicates that Crash-day observations are associated with markedly elevated ATR-14 and Bollinger bandwidth, and depressed OBV relative to Normal and High-Volatility days, consistent with the expected signature of a stress event.

To externally validate the per-stock, priority-based labeling scheme against known market-wide events, the cross-sectional proportion of stocks in the crash state was tracked over time and compared against two well-documented periods of DSE-wide stress: the 2010-2011 market crash and the COVID-19 crash of March-May 2020. As shown in Fig. 5, the smoothed 10-day rolling proportion of stocks in the crash state showed pronounced peaks coinciding with both crisis windows, providing face-validity evidence that per-stock crash labels aggregate coherently into market-wide stress signals despite being defined independently for each ticker.

A further validation check examined the 1,353 rows (0.146% of the final dataset, spanning 285 distinct tickers) flagged as suspected trading suspensions or corporate actions. Conditional on such an event, the probability of the following trading day being labeled Crash rose to 26.2%, against an unconditional base rate of approximately 5%, and the probability of Normal fell correspondingly to 49.3%. This substantial lift indicates that suspension events carry genuine predictive signal for subsequent market stress, supporting their retention in the dataset as informative observations rather than noise to be discarded.

Finally, the pooled fallback mechanism used for thin-history tickers was validated directly: the 11 tickers relying on pooled statistics exhibited a next-day crash rate of 4.11%, closely comparable to the 5.30% crash rate observed among tickers with sufficient own-history for individually fitted thresholds, confirming that the fallback mechanism produces labels of comparable quality for recently listed equities.

3.5 Model selection

We evaluate seven models spanning three methodological families: a linear baseline (Logistic Regression), tree-based ensembles (Random Forest, XGBoost), sequence models (LSTM, GRU), and two classical time-series models from financial econometrics (HMM, GJR-GARCH). All classification models are trained on the same feature set $\mathrm{x}_t \in \mathbb{R}^{18}$ to predict the three-class label $y_t \in \{0, 1, 2\}$ corresponding to Normal, High-Volatility, and Crash regimes, respectively, with class weights inversely proportional to class frequency applied during training to counteract the natural rarity of the Crash class. The two econometric models operate directly on the return series and are included as domain-standard baselines that do not require the engineered feature set. Below we give the mathematical formulation underlying each model as implemented in our pipeline.

3.5.1 Logistic regression

As a linear baseline, we use multinomial (softmax) logistic regression with an elastic-net penalty [59]. The model estimates class-conditional probabilities from a linear score for each class via the softmax function of Eq. 24, and the parameters are fit by minimizing the sample-weighted, elastic-net-regularized negative log-likelihood of Eq. 25 using the SAGA solver.

$$P(y_t = k \mid \mathrm{x}_t) = \frac{\exp(\mathrm{w}_k^\top \mathrm{x}_t + b_k)}{\sum_{j=0}^{2} \exp(\mathrm{w}_j^\top \mathrm{x}_t + b_j)}, \tag{24}$$

$$\mathscr{L}(\mathrm{W}) = -\sum_{t=1}^{N} s_t \sum_{k=0}^{2} \mathbb{1}[y_t = k] \log P(y_t = k \mid \mathrm{x}_t) + \lambda \left(\rho \|\mathrm{W}\|_1 + \frac{1-\rho}{2} \|\mathrm{W}\|_2^2 \right), \tag{25}$$

Here, for input $\mathrm{x}_t \in \mathbb{R}^d$ and class $k \in \{0, 1, 2\}$, $\mathrm{w}_k \in \mathbb{R}^d$ and b_k are the weight vector and bias for class k; s_t is the inverse-frequency sample weight for observation t; $\lambda = 1/C$ controls the overall regularization strength; and $\rho \in [0, 1]$ is the L1 mixing ratio, with $\rho = 0$, $\rho = 1$, and $0 < \rho < 1$ corresponding to pure L2, pure L1, and elastic-net penalties, respectively.

3.5.2 Random forest

Random Forest is an ensemble of B decision trees, each trained on a bootstrap sample of the training data with a random subset of $m \leq d$ features considered at each split, so as to decorrelate individual trees [60]. Each tree recursively partitions the feature space by selecting, at each node, the split that maximizes the weighted impurity decrease of Eq. 26, where impurity is measured by the sample-weighted Gini index of Eq. 27; the final class-probability estimate is obtained by averaging leaf-node class distributions across all trees as in Eq. 28.

$$(j^*, \tau^*) = \arg\max_{j,\tau} \left[H(\mathscr{D}) - \frac{|\mathscr{D}_L|}{|\mathscr{D}|} H(\mathscr{D}_L) - \frac{|\mathscr{D}_R|}{|\mathscr{D}|} H(\mathscr{D}_R) \right], \tag{26}$$

$$H(\mathscr{D}) = \sum_{k=0}^{2} \hat{p}_k(\mathscr{D}) \big(1 - \hat{p}_k(\mathscr{D})\big), \qquad \hat{p}_k(\mathscr{D}) = \frac{\sum_{i \in \mathscr{D}} s_i \mathbb{1}[y_i = k]}{\sum_{i \in \mathscr{D}} s_i}. \tag{27}$$

$$P(y_t = k \mid \mathrm{x}_t) = \frac{1}{B}\sum_{b=1}^{B} \hat{p}_k\big(\mathrm{leaf}_b(\mathrm{x}_t)\big), \tag{28}$$

Here, $\mathscr{D}$ is the sample set at the current node; $\mathscr{D}_L = \{i \in \mathscr{D} : x_{ij} \le \tau\}$ and $\mathscr{D}_R = \mathscr{D} \setminus \mathscr{D}_L$ are the resulting child partitions for candidate split (j, τ); $H(\cdot)$ is the sample-weighted Gini impurity, with $\hat{p}_k(\mathscr{D})$ the weighted proportion of class-*k* samples in $\mathscr{D}$ and s_i the sample weight of observation *i*; and $\mathrm{leaf}_b(\mathrm{x}_t)$ denotes the terminal node of tree T_b reached by input x_t, with *B* the total number of trees.

3.5.3 XGBoost

XGBoost is a gradient-boosted tree ensemble that builds an additive model $F(\mathrm{x}_t) = \sum_{m=1}^{M} f_m(\mathrm{x}_t)$, where each f_m is a regression tree [61], by minimizing the regularized objective in Eq. 29 at each boosting round. Using a second-order Taylor expansion of the loss, the optimal leaf weight and the corresponding split gain reduce to the closed-form expressions in Eq. 30, with subsample and column-subsample ratios applied stochastically at each round and a learning rate shrinking each tree's contribution.

$$\mathscr{L}^{(m)} = \sum_{t=1}^{N} s_t\, \ell\big(y_t, \hat{y}_t^{(m-1)} + f_m(\mathrm{x}_t)\big) + \Omega(f_m), \qquad \Omega(f_m) = \gamma T + \frac{1}{2}\lambda \sum_{l=1}^{T} w_l^2, \tag{29}$$

$$w_l^* = -\frac{\sum_{t\in I_l} s_t g_t}{\sum_{t\in I_l} s_t h_t + \lambda}, \qquad \mathrm{Gain} = \frac{1}{2}\left[\frac{\big(\sum_{t\in I_L} s_t g_t\big)^2}{\sum_{t\in I_L} s_t h_t + \lambda} + \frac{\big(\sum_{t\in I_R} s_t g_t\big)^2}{\sum_{t\in I_R} s_t h_t + \lambda} - \frac{\big(\sum_{t\in I} s_t g_t\big)^2}{\sum_{t\in I} s_t h_t + \lambda}\right] - \gamma, \tag{30}$$

Here, $\ell(\cdot,\cdot)$ is the multi-class softmax cross-entropy loss; $\hat{y}_t^{(m-1)}$ is the ensemble prediction after $m-1$ rounds; *T* is the number of leaves in tree f_m; w_l is the score at leaf *l*; γ and λ penalize tree complexity; $g_t = \partial_{\hat{y}}\ell$ and $h_t = \partial^2_{\hat{y}}\ell$ are the first- and second-order gradients of the loss with respect to the current prediction; I_L, I_R are the sample sets of the left and right child leaves for a candidate split, with parent set $I = I_L \cup I_R$; s_t is the sample weight of observation *t*; and η (in the update $F^{(m)} = F^{(m-1)} + \eta f_m$) is the learning rate shrinking each new tree's contribution.

3.5.4 LSTM

For the recurrent models, the input at time *t* is a length-*L* window of feature vectors $\{\mathrm{x}_{t-L+1}, \ldots, \mathrm{x}_t\}$. The Long Short-Term Memory (LSTM) cell [62] updates a hidden state and cell state through input, forget, and output gates as given in Eq. 31–Eq. 36. Our architecture stacks two LSTM layers with dropout applied between layers, and the final hidden state is passed through the softmax output layer of Eq. 24 to obtain $P(y_t = k \mid \mathrm{x}_{t-L+1:t})$, trained by minimizing the sample-weighted cross-entropy loss of Eq. 25 (without the elastic-net penalty term).

$$\mathrm{f}_t = \sigma\big(W_f \mathrm{x}_t + U_f \mathrm{h}_{t-1} + b_f\big), \tag{31}$$

$$\mathrm{i}_t = \sigma\big(W_i \mathrm{x}_t + U_i \mathrm{h}_{t-1} + b_i\big), \tag{32}$$

$$\mathrm{o}_t = \sigma\big(W_o \mathrm{x}_t + U_o \mathrm{h}_{t-1} + b_o\big), \tag{33}$$

$$\tilde{\mathrm{c}}_t = \tanh\big(W_c \mathrm{x}_t + U_c \mathrm{h}_{t-1} + b_c\big), \tag{34}$$

$$\mathrm{c}_t = \mathrm{f}_t \odot \mathrm{c}_{t-1} + \mathrm{i}_t \odot \tilde{\mathrm{c}}_t, \tag{35}$$

$$\mathrm{h}_t = \mathrm{o}_t \odot \tanh(\mathrm{c}_t), \tag{36}$$

Here, f_t, i_t, and o_t are the forget, input, and output gates, respectively; $\tilde{\mathrm{c}}_t$ is the candidate cell state; c_t and h_t are the cell state and hidden state; $\sigma(\cdot)$ is the logistic sigmoid; $\odot$ denotes elementwise multiplication; and W_*, U_*, b_* are the learned input-to-hidden weight matrices, hidden-to-hidden weight matrices, and biases associated with each gate.

3.5.5 GRU

The Gated Recurrent Unit (GRU) [63] simplifies the LSTM by merging the cell and hidden states and using two gates instead of three, as given in Eq. 37–Eq. 40. As with the LSTM, we stack two GRU layers with inter-layer dropout, and the final hidden state is mapped to class probabilities via the softmax layer of Eq. 24. The absence of a separate cell state and output gate gives the GRU fewer parameters than the LSTM for a given hidden size, consistent with its substantially lower tuning cost.

$$\mathrm{z}_t = \sigma\big(W_z \mathrm{x}_t + U_z \mathrm{h}_{t-1} + b_z\big), \tag{37}$$

$$\mathrm{r}_t = \sigma\big(W_r \mathrm{x}_t + U_r \mathrm{h}_{t-1} + b_r\big), \tag{38}$$

$$\tilde{\mathrm{h}}_t = \tanh\big(W_h \mathrm{x}_t + U_h(\mathrm{r}_t \odot \mathrm{h}_{t-1}) + b_h\big), \tag{39}$$

$$\mathrm{h}_t = (1 - \mathrm{z}_t) \odot \mathrm{h}_{t-1} + \mathrm{z}_t \odot \tilde{\mathrm{h}}_t, \tag{40}$$

Here, z_t and r_t are the update and reset gates, respectively; $\tilde{\mathrm{h}}_t$ is the candidate hidden state; h_t is the resulting hidden state; $\sigma(\cdot)$ is the logistic sigmoid; $\odot$ denotes elementwise multiplication; and W_*, U_*, b_* are the learned input-to-hidden weight matrices, hidden-to-hidden weight matrices, and biases associated with each gate.

3.5.6 Hidden Markov model (HMM)

The HMM models the observed return series $\{r_t\}$ as generated by an unobserved discrete-state Markov chain $\{S_t\}$, $S_t \in \{1, \ldots, K\}$, with K latent regimes assumed to correspond approximately to the Normal, High-Volatility, and Crash classes [64]. The model specification, given in Eq. 41, is fit via the Baum-Welch (EM) algorithm, which alternates computing posterior state probabilities using the forward-backward recursions of Eq. 42 with a closed-form M-step; at inference time, the crash-state posterior probability is used directly as the model's predicted crash probability.

$$\pi_i = P(S_1 = i), \qquad A_{ij} = P(S_t = j \mid S_{t-1} = i), \qquad r_t \mid S_t = i \sim \mathcal{N}(\mu_i, \sigma_i^2), \tag{41}$$

$$\alpha_t(i) = \Big[\sum_{j=1}^{K} \alpha_{t-1}(j) A_{ji}\Big]\, \mathcal{N}(r_t; \mu_i, \sigma_i^2), \qquad \beta_t(i) = \sum_{j=1}^{K} A_{ij}\, \mathcal{N}(r_{t+1}; \mu_j, \sigma_j^2)\, \beta_{t+1}(j), \tag{42}$$

Here, π_i is the initial probability of latent state i; A_{ij} is the transition probability from state i to state j; μ_i and σ_i^2 are the mean and variance of the Gaussian emission distribution for state i; $\theta = \{\boldsymbol{\pi}, A, \{\mu_i, \sigma_i^2\}_{i=1}^{K}\}$ collects all model parameters;

$\gamma_t(i) = P(S_t = i \mid r_{1:T}, \theta)$ is the posterior probability of being in state i at time t given the full return sequence; and $\alpha_t(i)$, $\beta_t(i)$ are the forward and backward variables used to compute $\gamma_t(i)$ and the pairwise transition posteriors.

3.5.7 GJR-GARCH

The GJR-GARCH(1, 1) model captures the leverage effect, whereby negative return shocks increase conditional volatility more than positive shocks of equal magnitude [65]. Conditional variance evolves according to Eq. 43, and parameters are estimated by maximizing the Gaussian quasi-log-likelihood of Eq. 44; a crash signal is then derived by thresholding the one-step-ahead conditional volatility forecast (or an associated tail probability) against a chosen quantile, providing a purely volatility-driven baseline that, unlike the other six models, uses no engineered features beyond the return series itself.

$$\sigma_t^2 = \omega + \left(\alpha + \gamma \mathbb{1}[\epsilon_{t-1} < 0]\right)\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{43}$$

$$\ell(\omega, \alpha, \gamma, \beta) = -\frac{1}{2}\sum_{t=1}^{T}\left[\log(2\pi) + \log\sigma_t^2 + \frac{\epsilon_t^2}{\sigma_t^2}\right]. \tag{44}$$

Here, returns are modeled as $r_t = \mu + \epsilon_t$ with $\epsilon_t = \sigma_t z_t$ and $z_t \sim \mathcal{N}(0, 1)$; σ_t^2 is the conditional variance at time t; $\omega > 0$, $\alpha, \beta \geq 0$ with $\alpha + \gamma \geq 0$ are the baseline variance, ARCH, and GARCH coefficients; and γ is the asymmetry (leverage) parameter capturing the additional volatility contribution of negative shocks, with $\gamma = 0$ recovering the standard symmetric GARCH(1, 1) model.

3.6 Model training

The final model set was designed to span three distinct methodological paradigms: linear, tree-ensemble, and sequential deep learning, supplemented by two classical time-series models, to provide a comprehensive comparison of algorithmic approaches to market-stress classification. Logistic Regression serves as an interpretable linear baseline; Random Forest and XGBoost represent bagging- and boosting-based tree ensembles respectively, offering complementary strengths in variance reduction and sequential error correction; LSTM and GRU networks introduce recurrent sequence modeling capable of capturing temporal dependencies across a rolling 20-day lookback window, directly addressing the need for architectures with demonstrated effectiveness in financial time-series classification; and a GJR-GARCH and a HMM were additionally fitted as classical econometric benchmarks explicitly modeling volatility clustering and latent regime-switching behavior [66, 67].

3.6.1 Validation strategy

All tree-ensemble, linear, and deep-learning models were validated using an expanding-window walk-forward cross-validation scheme, comprising five sequential folds fitted exclusively on the 2008–2018 training partition, with the 2019–2022 partition reserved as a final untouched test set. In each fold, the training window expands to include all preceding trading dates, while the validation window covers a forward-contiguous block of subsequent dates; a five-trading-day embargo period was additionally removed immediately preceding each validation window to reduce leakage arising from residual serial

correlation in rolling-window features and labels near the fold boundary. This design directly avoids the leakage risks associated with a single arbitrary hold-out split or a non-chronological k-fold scheme, ensuring that no model is ever validated on data temporally preceding its training window.

3.6.2 Class-imbalance handling

Given the substantial class imbalance documented in Table 3, two complementary strategies were evaluated in parallel for every applicable model: class-weighting, in which sample weights were recomputed independently within each cross-validation fold's training partition using the inverse class-frequency scheme, and the Synthetic Minority Oversampling Technique (SMOTE), applied strictly within an in-fold resampling pipeline such that oversampling was refit on each fold's training partition alone and never applied to validation or test data. This fold-local refitting was enforced for both strategies to prevent information from validation folds influencing the resampling or weighting scheme, and resampling ratios and synthetic sample counts were logged for transparency. Crash-class precision-recall area under the curve (PR-AUC) was adopted as the primary scoring metric for both hyperparameter selection and imbalance-strategy comparison, in preference to accuracy or ROC-AUC, both of which are known to present an overly optimistic picture of minority-class performance under severe imbalance.

3.6.3 Hyperparameter search strategy

An exhaustive grid search across the full hyperparameter space of each model was computationally infeasible given the dataset size (630,356 training observations across five folds). Hyperparameter tuning was therefore conducted as a staged, budget-bounded randomized search: for each tree-ensemble model, the search space was partitioned into four sequential stages, structural parameters, sampling parameters, regularization parameters, and the learning-rate, ensemble-size trade-off, with each stage exploring four candidate configurations across the five cross-validation folds (20 fits per stage, 80 fits in total per model), and each stage's selected configuration carried forward as the fixed baseline for the subsequent stage. This staged approach allows a broader hyperparameter space to be explored than a single exhaustive grid would permit within a fixed, explicitly stated computational budget. For Logistic Regression, whose hyperparameter space is comparatively small, a two-stage search was used: an initial search over the regularization strength C using the `lbfgs` solver with ℓ_2 regularization, followed by a comparison of penalty types (ℓ_1, ℓ_2, elastic-net) using the `saga` solver around the best-performing C value. For the LSTM and GRU architectures, a lighter single-validation-split search over four candidate configurations was used in place of full five-fold tuning, reflecting the substantially higher computational cost of training recurrent architectures at this data scale; the selected configuration was subsequently validated using the full five-fold walk-forward scheme to obtain the reported cross-validated performance estimates. The complete hyperparameter search space for the tree-ensemble and linear models is summarized in Table 5, and the candidate configurations evaluated for the recurrent deep learning architectures are summarized in Table 6.

Table 5 Staged hyperparameter search space for the tree-ensemble and linear models

Model	Stage	Hyperparameter	Search values
XGBoost	1 – Structure	max_depth	3, 5, 7, 9
		min_child_weight	1, 5, 10, 20
	2 – Sampling	subsample	0.7, 0.8, 0.9, 1.0
		colsample_bytree	0.7, 0.8, 0.9, 1.0
	3 – Regularization	gamma	0, 0.1, 0.5, 1.0
		reg_alpha	0, 0.1, 1.0, 2.0
		reg_lambda	1, 2, 5, 10
	4 – Learning rate / size	learning_rate	0.01, 0.03, 0.05, 0.1
		n_estimators	150, 300, 450, 600
Random forest	1 – Structure	max_depth	8, 15, 25, 35
		min_samples_split	2, 5, 10, 20
	2 – Sampling	max_features	sqrt, log2, 0.5, 0.8
		max_samples	0.5, 0.65, 0.8, 0.95
	3 – Leaf regularization	min_samples_leaf	1, 2, 5, 10
		ccp_alpha	0.0, 0.0001, 0.001, 0.005
	4 – Ensemble size	n_estimators	100, 150, 200, 250
Logistic regression	1 – Regularization strength	C	0.001, 0.01, 0.1, 1.0, 10.0, 100.0
		penalty	ℓ_2 (fixed)
		solver	lbfgs (fixed)
	2 – Penalty / solver	C	$0.5C^*, C^*, 2C^*$
		penalty, solver	ℓ_1/ℓ_2/elastic-net, saga

Each stage evaluated four candidate configurations (six for Logistic Regression Stage 1) across five expanding-window cross-validation folds, with the best-performing configuration from each stage carried forward as fixed for subsequent stages. C^* denotes the best regularization strength identified in Stage 1

Table 6 Candidate architecture configurations evaluated for the recurrent deep learning models via single-validation-split search, prior to full five-fold cross-validation of the selected configuration

Model	Hyperparameter	Candidate values (single-split search)
LSTM	LSTM units (layer 1/layer 2)	64/32 (fixed across candidates)
	Dropout rate	0.2, 0.3, 0.3, 0.4
	Learning rate	$1 \times 10^{-3}, 1 \times 10^{-3}, 5 \times 10^{-4}, 5 \times 10^{-4}$
	Batch size	256 (fixed)
GRU	GRU units (layer 1/layer 2)	64/32 (fixed across candidates)
	Dropout rate	0.2, 0.3, 0.3, 0.4
	Learning rate	$1 \times 10^{-3}, 1 \times 10^{-3}, 5 \times 10^{-4}, 5 \times 10^{-4}$
	Batch size	256 (fixed)

3.6.4 Classical time-series models

Two classical econometric models were trained independently for each ticker using only the training-period return series, providing benchmark model families that are fundamentally different from both the tree-based ensemble methods and the recurrent deep learning approaches employed in this study. A Gaussian Hidden Markov Model (HMM) with three latent states, representing the Normal, High-Volatility, and Crash market regimes, was estimated using the Baum-Welch expectation-maximization algorithm with a diagonal covariance structure. Model estimation was performed for a maximum of 200 iterations with a convergence tolerance of 1×10^{-4}. No hyperparameter optimization was conducted for the HMM because the number of latent states was fixed *a priori* to align with the predefined three-class market stress formulation rather than treated as a tunable parameter.

The GJR-GARCH model was incorporated as a volatility-based benchmark rather than as a native classifier, allowing the predictive capability of conditional volatility forecasting to be compared with the proposed classification frameworks. An asymmetric GJR-GARCH(1,1) model with Student-t distributed innovations was fitted independently for each ticker using maximum likelihood estimation to capture both volatility clustering and the leverage effect, whereby negative return shocks produce a greater increase in future conditional volatility than positive shocks of similar magnitude. The fitted model generates only a one-step-ahead conditional mean return and conditional volatility forecasts, denoted by $\hat{\mu} * t + 1$ and $\hat{\sigma} * t + 1$, respectively. To obtain discrete class predictions, these forecasts were transformed into the same three-class formulation used throughout the study by applying threshold-based decision rules consistent with the ground-truth labeling procedure. Specifically, a ticker-day was classified as Crash whenever the predicted conditional mean return satisfied $\hat{\mu} * t + 1 < \mu * \text{train} - 2\sigma_{\text{train}}$, where μ_{train} and σ_{train} denote the mean and standard deviation of the training-period returns. The High-Volatility class was identified whenever the predicted conditional volatility exceeded the 75th percentile of the ticker's in-sample fitted conditional volatility distribution, thereby providing a GARCH-native approximation that is threshold-equivalent to the intraday-range-based criterion used to construct the primary target labels. Tickers with insufficient training observations or fitted models violating the stationarity condition were excluded from the GARCH-based evaluation to ensure reliable parameter estimation and meaningful benchmark comparisons.

3.7 Feature importance

To improve interpretability while avoiding the known bias of impurity-based feature importance toward high-cardinality and frequently-split features, model interpretation was conducted using out-of-sample permutation importance and SHapley Additive exPlanations (SHAP) [68] rather than relying solely on each model's built-in importance mechanism. For the tree-ensemble models (Random Forest, XGBoost), SHAP values were computed using the tree-based explainer, which derives exact Shapley values efficiently from the model's internal tree structure; for Logistic Regression, coefficient magnitudes were retained as a directly interpretable measure of linear influence, and SHAP values were additionally computed using a linear explainer for consistency with the other models. For the LSTM and GRU architectures, for which standard permutation importance does not accommodate three-dimensional sequence input, a custom permutation procedure was implemented in which a given feature's values were shuffled across the sample axis while preserving the temporal structure of all other features and time steps, and the resulting degradation in Crash-class PR-AUC was recorded as that feature's importance. Across all models, permutation importance and SHAP values were computed on the held-out test set to reflect out-of-sample feature relevance rather than in-sample fitting behavior, and the two features most structurally proximate to the label-generating rule, daily return and intraday volatility, were evaluated separately via a companion model trained with these features excluded, to quantify their marginal contribution to crash detection; all resulting importance rankings are interpreted as associational rather than causal. Importance scores were computed independently for each model and compared collectively to identify features that ranked consistently highly

across the different algorithmic paradigms, providing a robust and model-agnostic interpretation of feature relevance underlying market-stress classification.

4 Results analysis

4.1 Baseline and deterministic-rule benchmarks

Before evaluating the full model set, a series of simple diagnostic benchmarks were established to determine whether the subsequent machine learning and time-series models provide genuine predictive value beyond trivial decision rules. The contemporaneous labeling rule itself, which defines `class_same_day` directly from same-day raw return and volatility values, recovers itself with 100% accuracy by construction; this result is reported purely as a sanity check confirming internal consistency of the label-generating rule, rather than as a competing forecasting model, since the genuine forecasting task requires predicting the *following* day's market-stress state.

Two naive forecasting baselines were evaluated on the held-out test set. A naive persistence rule, which predicts tomorrow's class as identical to today's realized class, achieved an overall accuracy of 0.7262, a Crash-class recall of 0.1737, and a Crash-class precision of 0.1736. A single-variable lagged-volatility heuristic, which flags a High-Volatility day whenever the previous day's intraday volatility exceeds its training-period 80th percentile, achieved an accuracy of 0.7447, though this rule does not attempt to distinguish the Crash class. Logistic Regression, which additionally serves as one of the five primary classifiers evaluated in this study, achieved a Crash-class PR-AUC of 0.1222 as a genuine statistical baseline incorporating the full engineered feature set. A GJR-GARCH(1,1) model was additionally fitted on the pooled market index as a preliminary volatility-based reference; the fitted mean-equation intercept ($\mu = -0.0184, p = 0.0227$) confirms a statistically significant negative average return consistent with the fat-tailed, mildly downward-drifting character of the pooled return series prior to per-ticker modeling, which is treated separately as a full baseline model in subsequent sections.

4.2 Classifier performance: cross-validation and held-out test

Table 7 reports the cross-validated performance of all seven models: Logistic Regression, Random Forest, XGBoost, LSTM, GRU, HMM, and GJR-GARCH, averaged across the five expanding-window cross-validation folds described in the Methods section, with Crash-class PR-AUC [69] adopted as the primary evaluation criterion given the severity of class imbalance. Random Forest achieved the highest cross-validated crash PR-AUC (0.1697 ± 0.0134), followed by XGBoost (0.1409 ± 0.0236) and LSTM (0.1407 ± 0.0164), while the two classical time-series models exhibited comparatively

Table 7 Cross-validated Crash-class and overall discrimination metrics (mean ± std across five expanding-window folds)

Model	Crash PR-AUC	Macro PR-AUC	Crash recall	Crash precision	ROC-AUC (OvR)
Logistic regression	0.1205 ± 0.0302	0.4595 ± 0.0048	0.2456 ± 0.0853	0.1356 ± 0.0318	0.724 ± 0.0253
Random forest	0.1697 ± 0.0134	0.4851 ± 0.0036	0.2433 ± 0.0405	0.1878 ± 0.0164	0.7446 ± 0.0235
XGBoost	0.1409 ± 0.0236	0.4693 ± 0.0046	0.2772 ± 0.0762	0.1456 ± 0.0325	0.735 ± 0.0239
LSTM	0.1407 ± 0.0164	0.4671 ± 0.0102	0.1976 ± 0.0518	0.1551 ± 0.0282	0.7322 ± 0.0429
GRU	0.1316 ± 0.0187	0.4655 ± 0.0077	0.2181 ± 0.0678	0.1476 ± 0.0269	0.7294 ± 0.0388
HMM	0.0776 ± 0.0409	–	0.2418 ± 0.1341	0.0816 ± 0.0514	0.5918 ± 0.0308
GJR-GARCH	0.1248 ± 0.0213	–	0.0010 ± 0.0022	0.0169 ± 0.0378	0.6977 ± 0.0220

Macro PR-AUC is not reported for HMM and GJR-GARCH, as neither model produces a directly comparable full three-class probability output under identical fold-level scoring

limited discrimination for the Crash class, with GJR-GARCH at 0.1248 ± 0.0213 and the Hidden Markov Model markedly weaker at 0.0776 ± 0.0409.

Table 8 reports the corresponding held-out test-set (2019–2022) classification performance. All classifiers achieved strong recall on the dominant Normal class (0.73–0.85) while exhibiting the expected precision degradation on the minority Crash class characteristic of severe imbalance. Random Forest achieved the strongest overall weighted F1-score (0.7042) and the highest Crash-class precision (0.1514) among the five primary classifiers, while XGBoost, LSTM, and GRU achieved comparable or marginally higher Crash-class recall. The GJR-GARCH model, despite competitive Crash PR-AUC under the primary ranking metric, produced zero Crash-class recall and precision at its default classification threshold, and the Hidden Markov Model exhibited the weakest overall discrimination across all three classes.

Table 8 Held-out test-set (2019–2022) classification performance across all evaluated models

Model	Class	Precision	Recall	F1-score	Support
Logistic regression	Normal	0.8606	0.7777	0.8170	217,365
	High-Vol	0.4228	0.4645	0.4426	63,344
	Crash	0.1250	0.2573	0.1683	13,881
	Macro Avg	0.4695	0.4998	0.4760	294,590
	Weighted Avg	0.7318	0.6858	0.7060	294,590
Random forest	Normal	0.8665	0.7587	0.8090	217,365
	High-Vol	0.4059	0.5249	0.4578	63,344
	Crash	0.1514	0.2436	0.1867	13,881
	Macro Avg	0.4746	0.5091	0.4845	294,590
	Weighted Avg	0.7337	0.6842	0.7042	294,590
XGBoost	Normal	0.8711	0.7297	0.7941	217,365
	High-Vol	0.4025	0.4981	0.4452	63,344
	Crash	0.1220	0.2999	0.1734	13,881
	Macro Avg	0.4652	0.5092	0.4709	294,590
	Weighted Avg	0.7350	0.6596	0.6899	294,590
LSTM	Normal	0.8693	0.7522	0.8065	217,245
	High-Vol	0.4067	0.4794	0.4401	63,293
	Crash	0.1341	0.3078	0.1868	13,855
	Macro Avg	0.4700	0.5131	0.4778	294,393
	Weighted Avg	0.7352	0.6726	0.6986	294,393
GRU	Normal	0.8650	0.7786	0.8196	217,245
	High-Vol	0.4005	0.3855	0.3928	63,293
	Crash	0.1420	0.3887	0.2081	13,855
	Macro Avg	0.4692	0.5176	0.4735	294,393
	Weighted Avg	0.7311	0.6758	0.6990	294,393
HMM	Normal	0.7679	0.7341	0.7506	217,315
	High-Vol	0.3631	0.2078	0.2643	63,317
	Crash	0.0579	0.2111	0.0908	13,844
	Macro Avg	0.3963	0.3843	0.3686	294,476
	Weighted Avg	0.6475	0.5963	0.6150	294,476
GJR-GARCH	Normal	0.8195	0.8493	0.8341	203,068
	High-Vol	0.4293	0.4620	0.4451	60,338
	Crash	0.0000	0.0000	0.0000	12,916
	Macro Avg	0.4163	0.4371	0.4264	276,322
	Weighted Avg	0.6960	0.7251	0.7102	276,322

Support totals vary slightly across models: LSTM and GRU support is reduced by the 20-day lookback window required to construct input sequences; HMM support reflects rows retained after causal forward filtering; GJR-GARCH support (276,322 of 294,590 total test rows, 93.8% coverage) reflects the exclusion of 11 tickers for which the model failed to converge or violated the stationarity condition

Figure 6 presents the Precision, Recall, and F1-score of all seven models across the three market-stress classes, with horizontal error bars indicating the 95% confidence intervals estimated from the standard deviation across the five expanding-window cross-validation folds. The figure complements Table 7 by visually highlighting both performance and variability. Substantial overlap between the confidence intervals of models such as XGBoost and LSTM on the Crash class suggests that their performance differences are not statistically distinguishable based on cross-validation variability, consistent with the Statistical Significance Analysis. The Normal and High-Volatility classes exhibit relatively narrow confidence intervals, indicating more stable estimates, whereas the Crash class shows wider intervals due to its smaller and more variable sample size across folds. HMM and GJR-GARCH are reported only for the Crash class, as fold-level metrics for the other classes were not computed under their respective evaluation frameworks.

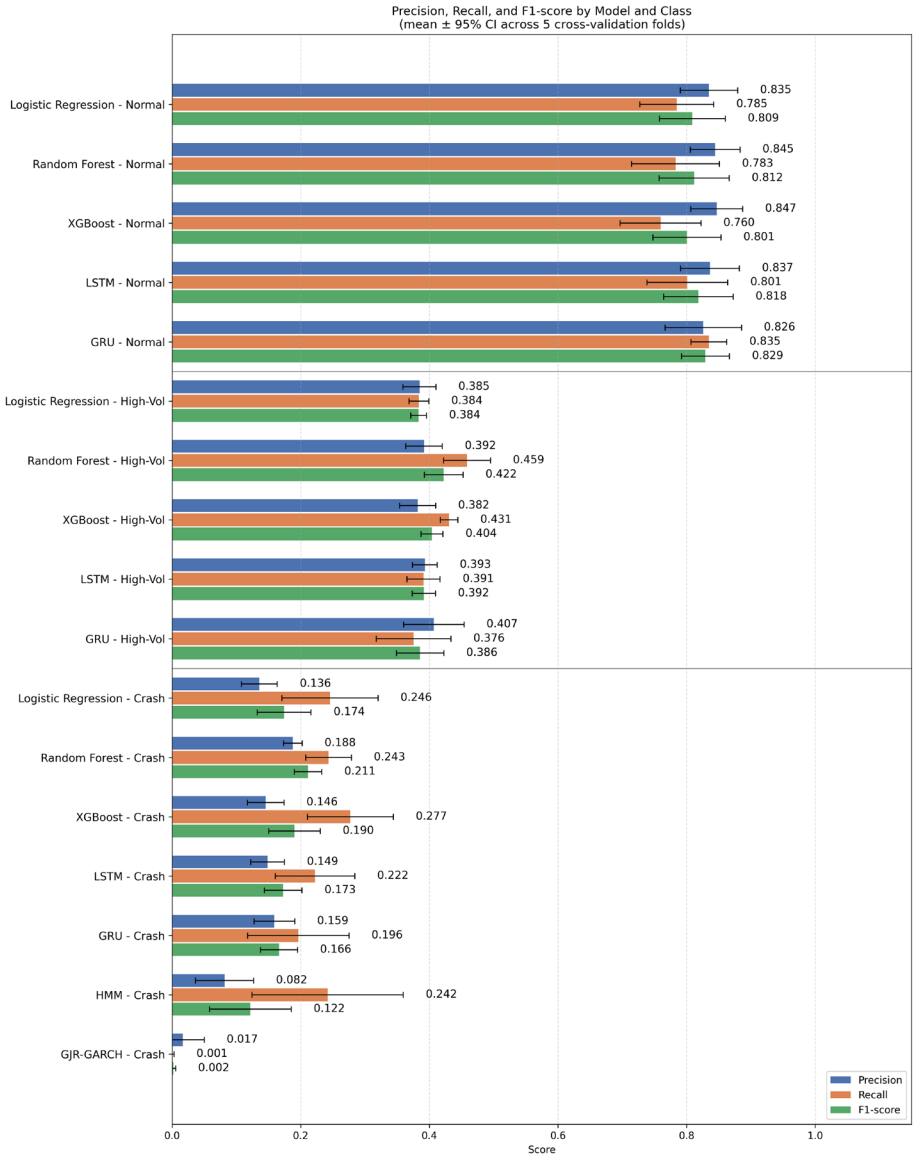


Fig. 6 Precision, Recall, and F1-score (mean ± 95% CI) by model and class across five cross-validation folds

4.3 Class-imbalance strategy: SMOTE versus class-weighting

For each of the five primary classifiers, class-weighting and SMOTE-based oversampling were evaluated as parallel strategies for addressing class imbalance, with SMOTE applied strictly within each cross-validation fold's training partition to prevent synthetic samples generated from validation-fold information. Table 9 summarizes the resulting Crash-class PR-AUC under both strategies. Class-weighting outperformed SMOTE for four of the five classifiers (Random Forest, Logistic Regression, LSTM, and GRU), while XGBoost showed a modest improvement under SMOTE (0.1268 vs. 0.1227). For an illustrative single fit on the full training partition, SMOTE increased the minority-class representation from an original distribution of 462,265 Normal, 127,738 High-Volatility, and 34,451 Crash observations to a fully balanced 462,265 observations per class. The comparatively limited benefit of SMOTE across most classifiers suggests that class-weighting, which adjusts the loss function directly rather than introducing synthetic interpolated observations, is generally the more effective strategy for this dataset, plausibly because SMOTE's linear interpolation between minority-class neighbors is less suited to the non-stationary, temporally-ordered structure of financial returns.

4.4 Precision-recall analysis and operational metrics

Given the severe rarity of the Crash class (approximately 5% of test-set observations), Crash-class precision-recall area under the curve (PR-AUC) is adopted as the primary discrimination metric throughout this study, in preference to accuracy or ROC-AUC [70], both of which are known to present an overly optimistic picture of minority-class performance under severe imbalance. Figure 7 presents the precision-recall curves for the Crash class across all seven models. Random Forest achieved the highest Crash-class average precision ($\mathrm{AP} = 0.1428$), followed closely by GJR-GARCH ($\mathrm{AP} = 0.1494$), GRU ($\mathrm{AP} = 0.1407$), LSTM ($\mathrm{AP} = 0.1361$), Logistic Regression ($\mathrm{AP} = 0.1270$), and XGBoost ($\mathrm{AP} = 0.1227$), while the Hidden Markov Model trailed substantially ($\mathrm{AP} = 0.0652$). All curves exhibit the characteristic steep early decay typical of severe class imbalance, in which precision at low recall is markedly higher than precision sustained across the full recall range.

Beyond the aggregate PR-AUC, Table 10 reports operational metrics directly relevant to a practical early-warning deployment: recall achievable at a fixed 5% false-positive-rate budget, precision among the model's most confident alerts (top-50 through top-1000), and event-level crash recall, which credits a detection if any alert falls within a one-trading-day tolerance window of a genuine crash episode rather than requiring exact row-level agreement. Under this operational framing, GRU achieved the highest event-level recall (0.5505), followed by Logistic Regression (0.5363) and LSTM (0.4890), while precision among the most confident alerts was substantially stronger for LSTM

Table 9 Comparison of class-weighting and SMOTE-based oversampling strategies, held-out test-set Crash-class PR-AUC

Model	Crash PR-AUC (class-weighted)	Crash PR-AUC (SMOTE)
Logistic regression	0.1270	0.1251
Random forest	0.1428	0.1390
XGBoost	0.1227	0.1268
LSTM	0.1361	0.1163
GRU	0.1407	0.1291

Both strategies are refit independently within each cross-validation fold's training partition

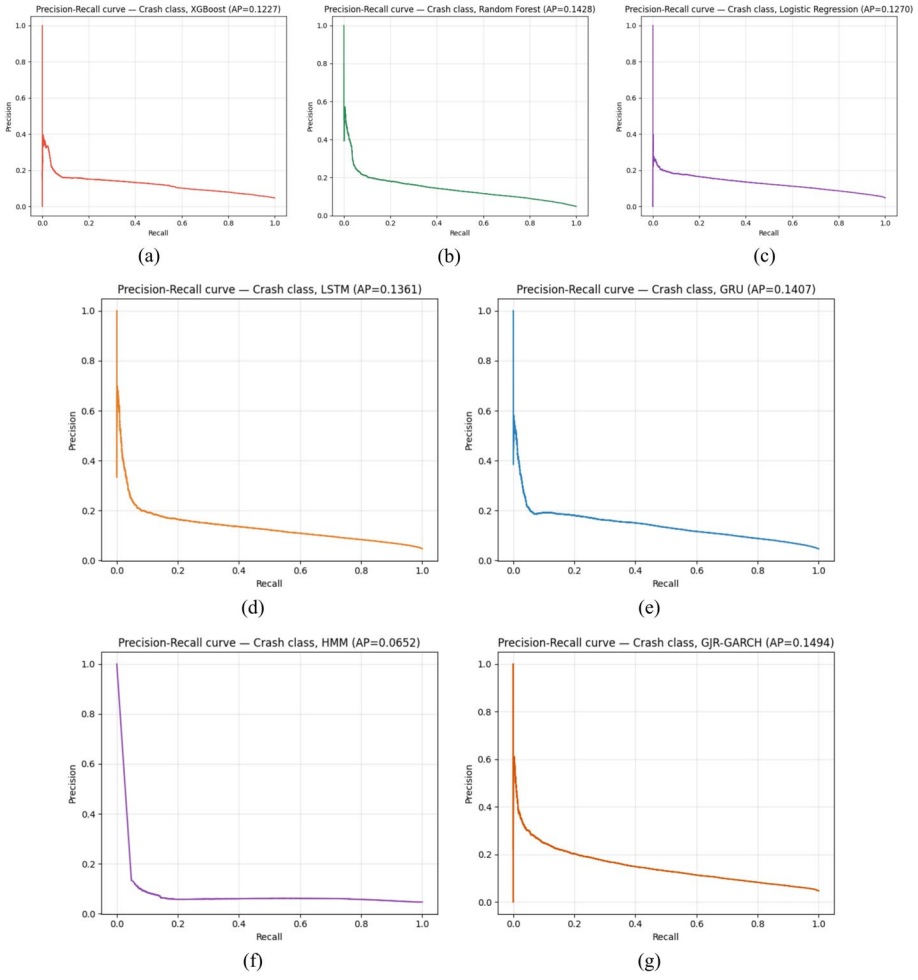


Fig. 7 Precision-recall curves for the Crash class on the held-out test set (2019-2022), shown for **a** XGBoost, **b** Random Forest, **c** Logistic Regression, **d** LSTM, **e** GRU, **f** HMM, and **g** GJR-GARCH. Average precision (AP), equivalent to Crash-class PR-AUC, is reported in each panel title. All models exhibit the characteristic steep early-recall precision decay associated with severe class imbalance (≈5% crash prevalence); the Hidden Markov Model (**f**) shows markedly weaker and flatter discrimination than the remaining six models

Table 10 Operational precision-recall metrics on the held-out test set

Model	Recall@FPR$\leq$5%	P@top-50	P@top-100	P@top-500	P@top-1000	Event-level recall
Logistic regression	0.1995	0.36	0.23	0.256	0.240	0.5363
Random forest	0.2186	0.50	0.52	0.438	0.385	0.3606
XGBoost	0.1832	0.26	0.36	0.352	0.329	0.4691
LSTM	0.2003	0.64	0.64	0.446	0.364	0.4890
GRU	0.2188	0.54	0.55	0.410	0.342	0.5505
HMM	0.0972	0.04	0.06	0.066	0.072	0.2567
GJR-GARCH	0.2271	0.00	0.00	0.000	0.044	0.0000

Event-level recall applies a ± 1 trading-day tolerance window around each distinct crash episode (approximately 10,700-11,470 episodes depending on model-specific test coverage), crediting detection if any alert falls within the window rather than requiring exact-day agreement

(0.64 at top-50 and top-100 alerts) than for any other model. GJR-GARCH and HMM, by contrast, achieved markedly weaker performance across all operational metrics, with GJR-GARCH producing zero precision among its top-50 through top-500 alerts and zero event-level recall, consistent with its zero row-level crash recall.

4.5 Calibration analysis

Table 11 reports calibration diagnostics for the Crash class across all seven models. Because the raw Brier score is close to automatic under a base rate of approximately 5% (a trivial classifier predicting the base rate everywhere already achieves a low absolute Brier score), the Brier Skill Score relative to a prevalence-only baseline is reported as the primary calibration metric, where a value of 0 indicates performance equivalent to the naive base-rate prediction and negative values indicate performance worse than this trivial baseline in terms of squared probability error. Notably, all seven models produced negative Brier Skill Scores, indicating that despite superior *ranking* performance relative to the prevalence baseline, the predicted probabilities themselves are not well calibrated in an absolute sense. This pattern is consistent with the effect of class-weighting, which is designed to improve the model's ability to *rank* minority-class observations correctly but systematically shifts predicted probabilities away from the true empirical base rate, degrading calibration even as discrimination improves. GJR-GARCH exhibited the calibration performance closest to the prevalence baseline (Brier Skill Score $= -0.0133$), reflecting its comparatively conservative probability outputs, while the Hidden Markov Model exhibited the poorest calibration by a substantial margin (Brier Skill Score $= -3.3143$), consistent with its comparatively concentrated, less-differentiated crash-state posterior probabilities.

Figure 8 presents the corresponding reliability diagrams. All models exhibit systematic underconfidence at higher predicted probabilities, with observed crash frequencies consistently falling below the predicted probability along the diagonal, most pronounced for XGBoost and least pronounced for GJR-GARCH, whose curve tracks closest to the perfectly-calibrated diagonal. The Hidden Markov Model's reliability curve is notably flat and non-monotonic, reflecting its comparatively coarse, three-state posterior probability structure.

4.6 Regime-specific robustness

To assess whether model performance is stable across distinct market conditions rather than an artifact of a particular sub-period, Crash-class PR-AUC was evaluated separately

Table 11 Crash-class calibration diagnostics on the held-out test set

Model	Brier score	Prevalence baseline	Brier skill score	ECE
Logistic regression	0.08317	0.04490	-0.8523	0.1841
Random forest	0.08286	0.04490	-0.8454	0.1863
XGBoost	0.10196	0.04490	-1.2708	0.2400
LSTM	0.08238	0.04485	-0.8369	0.1674
GRU	0.08314	0.04485	-0.8538	0.1703
HMM	0.19329	0.04480	-3.3143	–
GJR-GARCH	0.04515	0.04456	-0.0133	–

Brier Skill Score is computed relative to a prevalence-only (constant base-rate) baseline; 0 indicates performance equal to this baseline, negative values indicate worse absolute calibration despite potentially superior ranking performance. Expected Calibration Error (ECE) is omitted for HMM and GJR-GARCH, whose probability outputs are derived from fundamentally different mechanisms (state-posterior filtering and conditional-volatility thresholding, respectively) not directly comparable to the binned reliability calculation used for the classification models

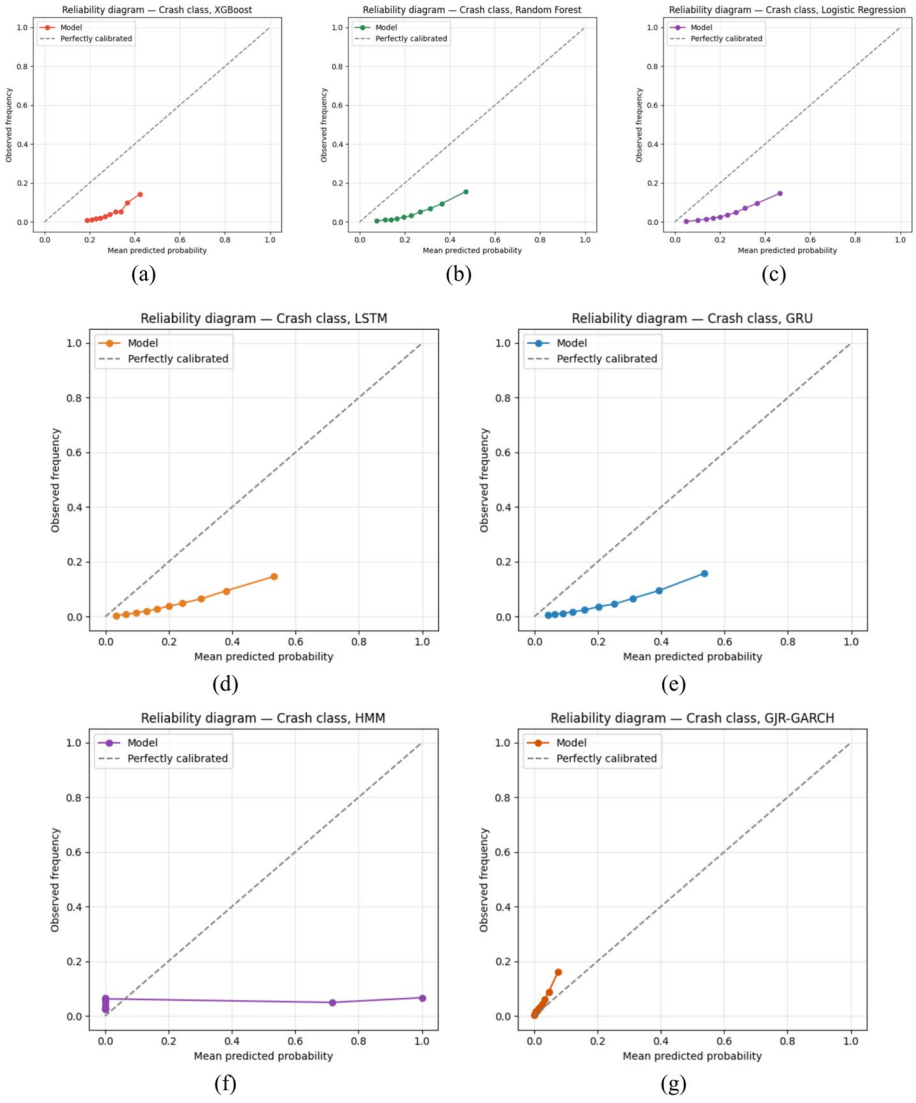


Fig. 8 Reliability diagrams for the Crash class on the held-out test set, shown for **a** XGBoost, **b** Random Forest, **c** Logistic Regression, **d** LSTM, **e** GRU, **f** HMM, and **g** GJR-GARCH. The dashed diagonal indicates perfect calibration. All models exhibit systematic underconfidence (curves below the diagonal), consistent with the negative Brier Skill Scores; GJR-GARCH (**g**) tracks closest to the diagonal, while HMM (**f**) exhibits a flat, non-monotonic calibration curve reflecting its coarser three-state posterior structure

within four regimes: the pre-COVID period (2019), the COVID-19 crash (March-May 2020), the post-COVID recovery period (June 2020-December 2021), and the Russia-Ukraine conflict period (February-December 2022). Table 12 reports crash PR-AUC by regime and model. Every model achieved its highest Crash-class PR-AUC during the COVID-19 crash window, with values roughly double those observed in the calmer surrounding regimes (for example, XGBoost rising from 0.1437 pre-COVID to 0.2257 during the COVID crash, and Logistic Regression from 0.1207 to 0.2864), consistent with the expectation that crash-detection signal strengthens during periods of genuine, concentrated market stress. Random Forest maintained the most consistent performance across all four regimes (0.1268–0.2453), while the Hidden Markov Model and GJR-GARCH showed comparatively larger degradation outside the COVID window,

Table 12 Regime-specific Crash-class PR-AUC

Regime	LR	RF	XGBoost	LSTM	GRU	HMM	GARCH
Pre-COVID (2019)	0.1207	0.1565	0.1437	0.1499	0.1472	0.0794	0.1133
COVID crash	0.2864	0.2453	0.2257	0.2594	0.2623	0.2317	0.2485
Post-COVID recovery	0.1125	0.1268	0.1057	0.1266	0.1320	0.0569	0.1274
Russia-Ukraine (2022)	0.1221	0.1277	0.0992	0.1071	0.1122	0.0447	0.1267

All models show substantially elevated crash-detection performance during the COVID-19 crash window relative to the surrounding calmer regimes, providing face-validity evidence that the models' Crash-class signal genuinely strengthens during periods of concentrated market stress rather than reflecting a fixed, period-independent decision rule

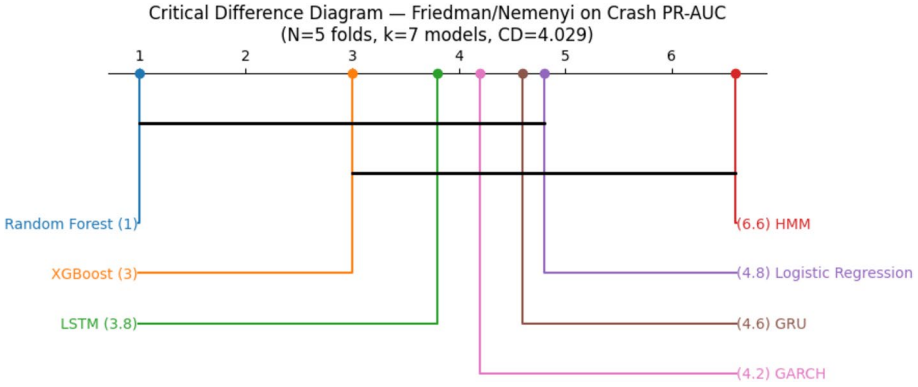


Fig. 9 Critical difference diagram of the Friedman/Nemenyi test on Crash-class PR-AUC across five expanding-window cross-validation folds and seven models ($N = 5$, $k = 7$, $CD = 4.029$). Models connected by a horizontal bar do not differ significantly at $\alpha = 0.05$. Random Forest achieves the best average rank (1.0); only its comparison against HMM (average rank 6.6) exceeds the critical difference

particularly during the Russia-Ukraine period (HMM: 0.0447; GJR-GARCH: 0.1267, closer to the ensemble average despite its otherwise weaker row-level discrimination).

4.7 Statistical significance analysis

A Friedman test [71] was conducted on the fold-level Crash-class PR-AUC values across all seven models and five expanding-window cross-validation folds ($N = 5$, $k = 7$), yielding a chi-square statistic of $\chi^2_F = 19.1143$ ($p = 0.0040$), which falls comfortably within the theoretical bound of $N(k-1) = 30$ for this configuration. The null hypothesis of equal model performance is therefore rejected at the 0.05 significance level, justifying post-hoc pairwise analysis. Average ranks (lower indicating better crash-PR-AUC performance) were: Random Forest (1.0), XGBoost (3.0), LSTM (3.8), GJR-GARCH (4.2), GRU (4.6), Logistic Regression (4.8), and HMM (6.6).

A Nemenyi post-hoc test yielded a critical difference of $CD = 4.0291$ at $\alpha = 0.05$. Figure 9 presents the resulting critical difference diagram. Only one pairwise rank difference exceeded the critical difference threshold: Random Forest versus HMM (rank difference $= 5.600$), indicating that Random Forest's superior crash-detection performance relative to the Hidden Markov Model is the only statistically distinguishable pairwise result at the fold-rank level, while all remaining pairwise comparisons, including Random Forest versus XGBoost, despite their differing average ranks, remain statistically indistinguishable given the available five folds.

Wilcoxon signed-rank tests were additionally computed for all 21 pairwise model comparisons using the same fold-level crash PR-AUC values, with Holm-Bonferroni correction applied for multiple comparisons. Prior to correction, the smallest attainable

two-sided p-value at $N = 5$ paired observations is 0.0625, achieved when one model outperforms another in all five folds; six of the 21 pairs achieved this minimum raw p-value (Random Forest versus each of Logistic Regression, XGBoost, LSTM, GRU, HMM, and GJR-GARCH). After Holm-Bonferroni correction across 21 comparisons, none of the pairwise differences remained statistically significant ($p_{\text{holm}} \geq 1.0$ for all pairs). This outcome, despite the significant omnibus Friedman result, reflects the substantially lower statistical power of pairwise post-hoc testing under strict multiple-comparison correction at this fold count, rather than a genuine absence of pairwise differences; the omnibus Friedman/Nemenyi analysis, which does not require correction across all 21 individual pairs, remains the more informative test of overall model differentiation given the available five-fold design. A McNemar's test, which would additionally require row-level paired test-set predictions rather than fold-aggregated performance metrics, was not computed for this reason and is noted as a direction for future work should row-level prediction logs be retained.

4.8 Feature importance and interpretability

Model interpretation was conducted using out-of-sample permutation importance and SHapley Additive exPlanations (SHAP) rather than relying on in-sample impurity-based importance, which is known to be biased toward high-cardinality and frequently-split features. Figure 10 presents SHAP summary plots for XGBoost, Random Forest, and Logistic Regression, and permutation importance bar charts for LSTM and GRU, for which standard SHAP computation was not pursued given the substantially higher computational cost of gradient-based explanation methods for recurrent sequence models.

Across the tree-ensemble models, Bollinger bandwidth (`bb_width`) and rolling 5-day volatility emerged as the dominant predictors: `bb_width` ranked first for XGBoost (mean importance $= 0.04433$) and second for Random Forest (0.01976), while rolling 5-day volatility ranked first for Random Forest (0.03018) and second for XGBoost (0.03075). Intraday volatility, one of the two features flagged as structurally proximate

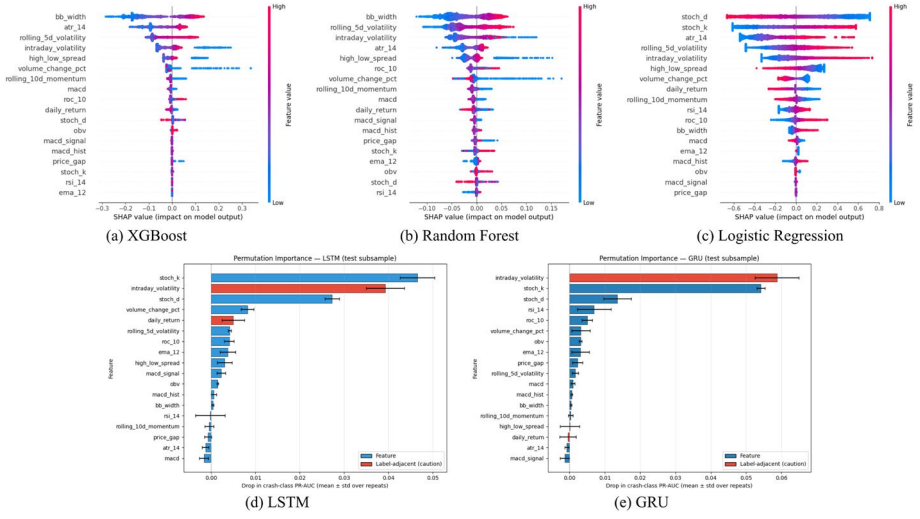


Fig. 10 Feature importance and interpretability results across five classifiers: SHAP summary plots for **a** XGBoost, **b** Random Forest, and **c** Logistic Regression, and out-of-sample permutation importance (mean ± std, Crash-class PR-AUC drop) for **d** LSTM and **e** GRU. Label-adjacent features (daily return, intraday volatility) are highlighted in red in panels (**d**)–(**e**). Bollinger bandwidth and rolling 5-day volatility dominate the tree-ensemble models, while GRU shows comparatively greater reliance on intraday volatility than the other four classifiers

Table 13 Marginal contribution of the two label-adjacent features (daily return, intraday volatility) to Crash-class PR-AUC, measured by comparing the full model against a companion model trained with these two features excluded

Model	Full model crash PR-AUC	Excl. label-adjacent features	Relative change
XGBoost	0.1227	0.1168	-4.8%
Random forest	0.1428	0.1348	-5.6%
Logistic regression	0.1270	0.1117	-12.0%

Not computed for LSTM, GRU, for which only permutation importance was obtained

Table 14 Fitted HMM transition matrix (training data)

	State 0	State 1	State 2
State 0	0.9407	0.0493	0.0099
State 1	0.0734	0.9046	0.0220
State 2	0.0042	0.0087	0.9872

Rows indicate the current state; columns indicate the transition probability to each subsequent state

to the label-generating rule, ranked third for both models (0.01039 and 0.01035, respectively), a moderate but not dominant contribution. For Logistic Regression, the stochastic oscillator components (`stoch_d`, `stoch_k`) and Average True Range emerged as the strongest linear predictors, with intraday volatility ranking fourth (0.03799).

For the recurrent architectures, permutation importance revealed a notable divergence in reliance on label-adjacent features. For LSTM, the stochastic oscillator (`stoch_k`, importance $= 0.04654$) ranked highest, with intraday volatility second (0.03929) and daily return considerably lower (0.00498, fifth-ranked). For GRU, by contrast, intraday volatility itself ranked as the single most important feature (0.05873), exceeding every other predictor including `stoch_k` (0.05419), indicating that GRU's crash-detection signal relies comparatively more heavily on a feature structurally close to the label-generating rule than any of the other four classifiers, a distinction worth noting when interpreting GRU's feature-attribution results as associational rather than causal.

To further quantify the marginal contribution of the two label-adjacent features (daily return and intraday volatility), a companion model was trained for each of the three interpretable classifiers with these two features excluded. Table 13 reports the resulting crash PR-AUC change. Excluding these features reduced crash PR-AUC by 4.8% for XGBoost, 5.6% for Random Forest, and 12.0% for Logistic Regression, indicating that while daily return and intraday volatility contribute measurably to crash detection, the majority of each model's discriminative signal derives from the remaining sixteen engineered features. All importance and contribution results reported in this subsection are interpreted as associational rather than causal.

4.9 Classical time-series models: parameter and state interpretation

4.9.1 Hidden Markov model

The three-state Gaussian Hidden Markov Model, fitted via Baum-Welch expectation-maximization on training-period data with daily return and intraday volatility excluded from the emission features to avoid label-adjacent leakage, converged successfully (final log-likelihood $= -9{,}168{,}373.95$). The fitted transition matrix (Table 14) reveals substantial state persistence, with self-transition probabilities of 0.9407, 0.9046, and 0.9872 for States 0, 1, and 2, respectively, corresponding to expected state durations of approximately 16.9, 10.5, and 77.9 trading days.

Table 15 Enrichment ratios (state-conditional class proportion divided by overall training-set base rate) used for state-to-label assignment

State	Normal	High-Vol	Crash
State 0 (assigned: Crash)	0.951	1.045	**1.487**
State 1 (assigned: High-Vol)	0.798	**1.578**	1.553
State 2 (assigned: Normal)	**1.100**	0.764	0.539

Bold values indicate the assigned class for each state. State 1's assignment reflects a narrow margin (1.578 vs. 1.553) between the High-Volatility and Crash candidate labels

Table 16 Standardized state-conditional feature means from the fitted Gaussian HMM (training partition)

Feature (standardized)	State 0 (crash)	State 1 (high-vol)	State 2 (normal)
High-low spread	0.672	0.801	−0.651
Volume change %	−0.038	−0.029	0.031
Price gap	0.033	0.267	−0.115
Rolling 5-day volatility	0.063	0.458	−0.202
Rolling 10-day momentum	−0.844	1.323	−0.038
EMA-12	0.514	0.482	−0.450
MACD	−0.507	0.810	−0.029
MACD signal	−0.269	0.347	0.015
MACD histogram	−0.621	1.002	−0.039
RSI-14	−0.522	1.050	−0.109
Stochastic %K	−0.359	0.932	−0.152
Stochastic %D	−0.375	0.971	−0.158
ROC-10	−0.599	1.152	−0.106
Bollinger bandwidth	0.264	0.458	−0.309
ATR-14	0.817	0.750	−0.709
OBV	−0.278	0.159	0.089

The state assigned to High-Volatility shows uniformly elevated momentum/oscillator readings, while the state assigned to Crash shows negative momentum alongside elevated volatility-range indicators (ATR, Bollinger bandwidth)

A majority-vote mapping of decoded training-set states to market-stress labels assigned all three hidden states to the Normal class, since Normal observations dominate every state in absolute terms under the prevailing 74/21/5 class distribution, an outcome that would render the HMM incapable of detecting either High-Volatility or Crash conditions. An enrichment-based mapping was therefore adopted instead, assigning each state to the class for which its observed proportion most exceeds that class's overall base rate (Table 15). Under this scheme, State 0 was mapped to Crash (enrichment ratio $= 1.487\times$ over its 5.53% training base rate), State 2 to Normal (enrichment $= 1.100\times$), and State 1 to High-Volatility (enrichment $= 1.578\times$); the assignment for State 1 is noted as comparatively low-confidence, given its narrow margin over the second-best candidate class (Crash, enrichment $= 1.553\times$, a margin of only 0.025).

Examination of the standardized state means (Table 16) supports this interpretation: the state assigned to High-Volatility (State 1) exhibits the highest values across nearly all momentum and oscillator-based features (MACD $= 0.810$, RSI-14 $= 1.050$, stochastic %K $= 0.932$), consistent with elevated but directionally ambiguous price activity, while the state assigned to Crash (State 0) exhibits negative momentum and oscillator readings alongside elevated ATR and Bollinger bandwidth, consistent with sharp downward price movement accompanied by expanded trading ranges.

To assess whether the HMM provides genuine predictive value, its performance was compared against two trivial baselines (Table 17): an always-Normal classifier and a

Table 17 HMM performance versus trivial baselines on the held-out test set

Model	Macro F1	Crash recall	Crash precision
Always-normal	0.2831	0.0000	–
Random (base-rate weighted)	0.3319	0.0526	0.0447
HMM (enrichment-based mapping)	0.3686	0.2111	0.0579

Table 18 Distribution of fitted GJR-GARCH(1,1) parameters across 300 converged equity tickers

	α	γ	β	Persistence	ν
Mean	0.1382	-0.0129	0.8028	0.9345	321.03
Std	0.0560	0.0515	0.1059	0.0845	120.65
Min	0.0043	-0.5107	0.0000	0.2190	2.43
25%	0.1103	-0.0275	0.7809	0.9313	228.93
Median	0.1326	-0.0091	0.8213	0.9502	316.84
75%	0.1553	0.0080	0.8528	0.9658	413.05
Max	0.7189	0.1842	0.9945	1.0000	500.00

α: shock sensitivity; γ: leverage/asymmetry parameter; β: volatility persistence coefficient; Persistence: $\alpha + \gamma/2 + \beta$; ν : Student-*t* degrees of freedom

random classifier weighted by the training-set class base rates. The fitted HMM (macro-F1 $= 0.3686$, crash recall $= 0.2111$, crash precision $= 0.0579$) outperformed both trivial baselines on macro-F1 and, critically, achieved non-trivial crash recall where the always-Normal baseline achieves none by construction; however, its absolute performance remained well below that of the supervised classifiers reported in Section 2, reflecting the inherent limitation of an unsupervised three-state regime model relative to models trained directly against the market-stress labels.

4.9.2 GJR-GARCH

An asymmetric GJR-GARCH(1,1) model with Student-*t* innovations was fitted independently for each of 311 equity tickers using training-period returns only. The model converged successfully for 300 tickers (96.5%), with 11 tickers excluded due to insufficient training observations or violation of the stationarity condition ($\alpha + \gamma/2 + \beta \geq 1$); the resulting test-set coverage was 276,322 of 294,476 rows (93.8%). Table 18 summarizes the distribution of fitted parameters across all 300 converged tickers. The mean estimated volatility persistence ($\alpha + \gamma/2 + \beta = 0.9345$) indicates substantial but stationary volatility clustering, consistent with well-documented behavior in equity return series. The leverage parameter γ was positive (indicating a genuine asymmetric response to negative shocks) for 34.7% of fitted tickers, with a ticker-level range from -0.511 to 0.184, indicating that while the classical leverage effect is present for a meaningful minority of DSE equities, it is not a universal feature of this market. The mean estimated Student-*t* degrees of freedom (321.03) suggests that, on average, fitted return distributions approach Gaussian tail behavior, though the minimum estimated value (2.43) indicates that a subset of tickers exhibit substantially fatter tails consistent with heavy-tailed return behavior.

The GJR-GARCH model's predicted test-set class distribution (76.17% Normal, 23.50% High-Volatility, 0.34% Crash) substantially under-predicts the Crash class relative to its true 4.68% prevalence in the covered test subset, consistent with the zero row-level crash recall; this reflects the model's reliance on a fixed mean/standard-deviation threshold applied to its own in-sample fitted conditional volatility, which under-detects the

rarest and most extreme tail events despite the model's respectable ranking-based crash PR-AUC.

4.10 Trade simulation and backtesting

To assess whether the crash-probability estimates translate into economic value beyond classification accuracy, we implement a simple risk-off trading rule on top of the Random Forest model's out-of-sample predictions. Using the Crash-class probability $\hat{p}_{\text{crash}}$ produced for each stock-day in the 2019–2022 test period, the strategy exits to cash (position $= 0$) whenever $\hat{p}_{\text{crash}}$ exceeds its 95th percentile ($\hat{p}_{\text{crash}} > 0.2276$) and otherwise holds a fully invested long position, consistent with the precision-at-top-k-alerts framing used elsewhere in our operational evaluation; this places the strategy in cash on only 5.00% of stock-days, so performance differences are driven by a small number of high-confidence crash alerts rather than by wholesale de-risking. Positions decided at day t are applied to the realized return at $t + 1$ to respect the label shift and avoid look-ahead bias, and returns are reconstructed from the Winsorized (capped) `daily_return` feature, a conservative choice that likely understates the strategy's true edge during extreme moves. On the pooled equal-weighted test set, the RF crash-gated strategy achieves a total return of 67.26% (15.01% annualized) versus 45.52% (10.74% annualized) for buy-and-hold, while simultaneously reducing annualized volatility (16.39% vs. 18.11%) and maximum drawdown (-32.21% vs. -38.41%), as shown in Figs. 11, 12. This translates into a higher Sharpe ratio (0.94 vs. 0.65) and Sortino ratio (1.30 vs. 0.91), with the drawdown plot showing that most of the strategy's advantage over buy-and-hold is concentrated in the COVID-19 crash window, where the crash-gate keeps the strategy shallower and lets it recover faster. Win rate (54.80% vs. 54.15%) and profit factor (1.18 vs. 1.13) are only marginally improved, indicating that the strategy's benefit comes primarily from avoiding a small number of large negative days rather than from systematically better day-to-day trade selection.

4.11 Computational cost and runtime

Beyond predictive performance, practical deployment of a crash-warning system depends on the computational budget required to train and tune it. Table 19 summarizes

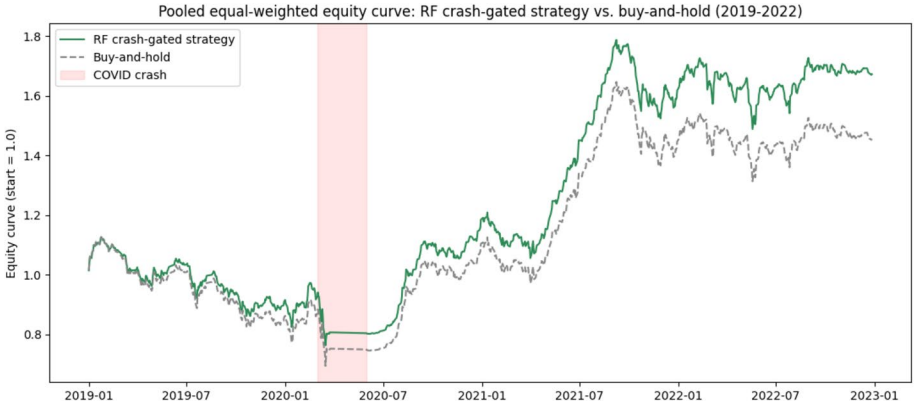


Fig. 11 Pooled equal-weighted equity curve (2019–2022, start $= 1.0$) for the Random Forest crash-gated strategy versus buy-and-hold, with the COVID-19 crash window shaded. The crash-gated strategy tracks buy-and-hold closely outside the crash window but avoids a large share of the COVID-19 drawdown, compounding to a higher terminal value

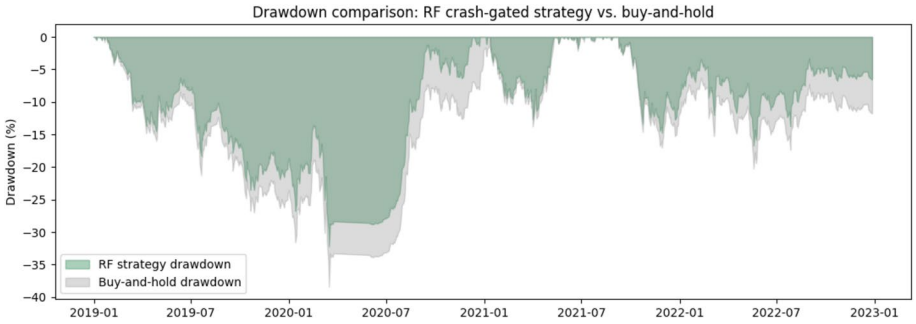


Fig. 12 Drawdown comparison between the Random Forest crash-gated strategy and buy-and-hold over the 2019–2022 test period. The crash-gated strategy consistently exhibits shallower drawdowns, most notably during the COVID-19 crash (2020 Q1-Q2), where its maximum drawdown (-32.21%) remains well above buy-and-hold's (-38.41%)

Table 19 Hyperparameter-tuning computational cost by model, aggregated across all staged grid-search rounds

Model	Total tuning time (min)	Total fits	Avg. time/fit (s)
Logistic regression	61.83	50	74.20
Random forest	160.38	80	120.29
XGBoost	38.83	80	29.13
LSTM	130.72	4	1960.75
GRU	10.52	4	157.75

Total Fits counts individual model fits (candidates × CV folds for Logistic Regression/Random Forest/XGBoost; candidates only for LSTM/GRU, which used a single fixed validation split per candidate rather than cross-validation)

the hyperparameter-search cost for each classifier, measured as wall-clock time on our Google Colab training environment, aggregated over the staged grid-search procedure. The tree ensembles and logistic regression were tuned via 5-fold cross-validated grid search (4–6 candidates per stage, 5 folds each), whereas the recurrent models were evaluated once per candidate on a fixed validation split, so their reported cost reflects per-candidate rather than per-fold training time. Random Forest is by far the most expensive model to tune, requiring over 160 min across 80 total fits (120.3 s/fit on average), roughly $4.1\times$ the cost of the class-weighted Logistic Regression baseline and $2.6\times$ the tuning cost of XGBoost (38.8 min, 29.1 s/fit, the cheapest per-fit cost of all five models). Among the recurrent architectures, GRU is markedly cheaper than LSTM (10.5 vs. 130.7 min total, roughly $12\times$ faster), reflecting GRU's simpler gating structure and correspondingly lower training cost. Overall, tuning cost varies by more than an order of magnitude across models (from 10.5 to 160.4 min), a difference worth weighing explicitly for any operational retraining pipeline, particularly one that would need to refit periodically as new market data arrives.

5 Discussion

Machine learning and deep learning techniques have been widely applied for stock market prediction using diverse indicators and modeling strategies. In India and Malaysia, SMA, EMA, and price-volume indicators combined with SVM, Random Forest, and LSTM improved risk-adjusted returns, although their effectiveness depended on market liquidity and trading volume [46, 47]. In China, cross-market features integrated with GRU and Transformer models enhanced prediction accuracy by capturing inter-market dependencies [50], while in South Korea, tree-based ensembles and deep learning

Table 20 Country-wise comparison of stock market stress prediction approaches

References	Country	Data	Features	Key insights
[46, 47]	India, Malaysia	Historical stock data	Trend-following, momentum, and volume-based technical indicators with SVM, Random Forest, LSTM	LSTM highest accuracy; better performance with high liquidity
[50]	China	Stock, bond, FX	Cross-market features with LSTM, GRU, Transformer	Cross-market integration improves accuracy; limited interpretability
[22]	South Korea	2812 stocks (2015–2024)	Implied volatility and wavelet price features with CatBoost, XGBoost, LGBM, CNN, LSTM, GRU, Transformer	Tree-based models excel; SHAP identifies key predictors
[51]	USA	Market VIX	Commodities, energy, policy, and geopolitical risks with multi-model transfer function optimized with PSO	Captures nonlinear effects; outperforms single models
Our study	Bangladesh (Dhaka)	DSE data (2008–2022)	Engineered market features (daily return, high-low spread, volume change, volatility, price gap etc) with machine learning models	Captures global feature importance; tailored to low liquidity and High-Volatility; interpretable predictions

Features are described concisely, highlighting key insights for each study

models leveraged implied volatility and wavelet-transformed features to predict market crashes, with SHAP analysis improving model interpretability [22]. Similarly, in the United States, multi-model transfer functions optimized using PSO effectively captured the nonlinear effects of commodities, energy markets, and geopolitical risks on market volatility [51]. Building on these studies, the proposed framework extends market stress prediction to the Dhaka Stock Exchange (DSE) by introducing a leakage-safe learning pipeline with robust statistical labeling, comprehensive feature engineering, chronological validation, and multiple machine learning and time-series models. The integration of feature importance analysis further improves model transparency by identifying the indicators that contribute most significantly to distinguishing Normal, High-Volatility, and Crash market conditions.

From a practical perspective, the proposed framework can serve as an early warning system for frontier markets such as the DSE. Investors and portfolio managers can utilize predicted market stress levels to support portfolio rebalancing, risk management, and hedging decisions, while regulators and policymakers can monitor evolving market conditions to identify potential systemic risks and implement timely interventions. The inclusion of industry-wise risk profiling further enables sector-level monitoring and supports informed decision-making during periods of heightened market uncertainty. Owing to its reliance on historical market data and computationally efficient models, the framework can be readily integrated into existing market surveillance and financial risk management systems. A country-wise comparison of stock market stress prediction approaches is summarized in Table 20.

Author contributions

Md Sadman Haque conceptualized and directed the research, overseeing data collection and preprocessing, designing the methodology, and authoring the original manuscript. Mohammad Sameer Ahmed contributed to the methodological development, carried out formal analysis, and participated in drafting and reviewing the manuscript. Md Tasfikur Rahman was involved in data analysis and feature engineering, and took part in writing, reviewing, and editing the manuscript. Robiul Awoul Robin played a key role in the data preprocessing and feature engineering phases. Zannatul Zahan Meem assisted in writing and reviewing the original draft of the manuscript. Jannatun Noor supervised the overall research and contributed to writing and reviewing the original draft.

Data availability

The data used in this study are publicly available on Kaggle. The dataset, titled Dhaka Stock Exchange, was published in 2023 and can be accessed at (https://www.kaggle.com/dsv/5210441).

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